

Bank Dividend Restrictions and Banks' Institutional Investors

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Abstract

Exploiting the 2020 Eurozone supervisory bank dividend restrictions, I use a within-investor difference-in-differences design to study the institutional investors' response. Netting out demand shocks, the average fund's ownership share proportionally declines by 19.5% by the end of the initial restriction, showing funds' active divestment. This reduction is driven by dividend-oriented funds and persists after restrictions are lifted. Pre-intervention data supports this clientele effect. Consistent with this clientele's pricing impact, restricted banks with greater ex ante fund exposure suffer significantly lower abnormal returns, highlighting how ownership structure shapes the valuation costs of supervisory interventions.

JEL Codes: G12, G21, G23, G28, G35

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1 Introduction

The entitlement to receive dividends from the company’s earnings typically resides in the hands of its shareholders and management. This can result in companies with a high propensity to pay dividends, for example banks¹, attracting investors who specifically seek out these cash flows. Recent evidence shows that among them are institutional investors, notably mutual funds (see Larkin, Leary and Michaely (2017) and Harris, Hartzmark and Solomon (2015)), a significant investor group with respect to the amount of stocks held². But in addition to shareholders and management, in 2020, a third party gained significance for the banking sector’s dividend payments – namely, the supervisory authority. Due to the COVID-19 pandemic, supervisors around the world imposed sector-wide dividend restrictions for banks, so as to increase their resilience by retaining profits instead of distributing them via dividends (Hardy, 2021). This interference in the payout policy was unprecedented by investors as it applied to all banks regardless of their performance or capitalization³. Due to the ad hoc nature of the decision, it was unclear how institutional investors of banks would react to this policy, as banks usually keep their dividend payments even during times of crisis (Cziraki, Laux and Lóránth, 2024; Acharya et al., 2022; Acharya, Le and Shin, 2017; Floyd, Li and Skinner, 2015; ?).

The change in policy poses the following research question, which I will analyze empirically in this paper: how do institutional investors of banks’ stock react to exogenously imposed temporary dividend restrictions? If there are institutional investors among banks’ shareholders who insist on receiving dividends, then the restrictions could lead to reductions in their ownership shares, as they might reallocate their investments towards companies that distribute dividends. Consequently, dividend restrictions could also have negative implications for banks’ equity valuations, potentially offsetting part of the intended resilience benefits of retained capital. Conversely, if these restrictions are perceived as a temporary measure, this could also lead to the inaction of investors, since portfolio

1. See Floyd, Li and Skinner (2015) and Gambacorta, Oliviero and Shin (2023) who show that banks have a higher propensity to pay dividends. Furthermore, Koussis and Makrominas (2019) show that banks also smooth their dividends.

2. For example Lambert, Vivar and Wedow (2024) show that in Europe institutional investors hold more than twice as many stocks than households.

3. Svoronos and Vrbaski (2020) give an overview of the BASEL III related dividend restrictions that are linked to banks’ capitalization, including capital buffers and the maximum distribution amount that supervisors can influence. Additionally, Matyunina and Ongena (2022) also discuss bank capital related payout restrictions and the newly imposed restrictions during the crisis. They note that these restrictions were unprecedented to investors.

adjustments in crisis periods can be very costly. Thus, investors might just choose to ride out the restriction period and not adjust their ownership shares. Which of the two mechanisms is at play needs to be tested empirically.

I address this question by analyzing the temporary dividend restrictions imposed in the Eurozone by the Single Supervisory Mechanism in light of the COVID-19 pandemic. Using monthly investor-level data and a difference-in-differences setup, I find that the average fund decreases its ownership shares in affected banks after the announcement of the dividend restrictions, in contrast to a comparable sample of unaffected Swiss banks. Since this measure was implemented during a period in which there were larger fund outflows, which could affect funds' ownership shares (Pástor and Vorsatz, 2020), I rely on an identification strategy that uses bank and also *investor* $\times$ *time* fixed effects. This allows me to control for time-varying investor demand factors, such as outflow-driven adjustments and helps me to pin down the effect by focusing on the within-investor-time variation in the data. Because the empirical design nets out these broad demand shocks, the results capture an active, targeted divestment decision by fund managers rather than a mechanical byproduct of market-wide selling. This targeted reduction is economically meaningful – the average fund's ownership share in treated banks proportionally declines by 19.5% by the end of the initial restriction period in November 2020 – and highly persistent, as fund investors do not revert to their previous ownership levels even after the payout restrictions are completely lifted.

At first glance, this appears inconsistent with a straightforward demand for dividends, as one might expect funds to return once payouts resume. However, temporary restrictions may increase uncertainty about future policy interventions and therefore about future dividend payments, as emphasized by Matyunina and Ongena (2022), reducing the attractiveness of bank equity for dividend-oriented investors. Moreover, frequent adjustments would increase transaction cost. This can alter the investment case for bank equity among income-oriented institutional investors⁴. Indeed, the European Systemic Risk Board (ESRB) later proposed in a March 2022 concept note that this tool should be used in crisis times in the future (ESRB, 2022). Therefore, the findings also show how even short-term interventions can have longer-term impacts through increased policy uncertainty.

4. "Equity investors have traditionally viewed banks as solid, if low growth businesses that can be relied upon to pay steady income. 'But dividend bans means that way of thinking does not really work any more,' said Mr Ramos-Martin [a global equities manager at Aviva Investors].", Walker, Owen. 2020. "Europe's banks fear investor flight after dividend bans." Financial Times, December 26. <https://www.ft.com/content/fd5eb169-ca1c-43c1-b67e-cbc7ede9988a>.

The driver behind funds' demand for dividends could be the desire to either attain a smooth cash flow via smooth dividends as Larkin, Leary and Michaely (2017) have documented, or to receive just a high dividend yield that they can subsequently distribute to their investors as Harris, Hartzmark and Solomon (2015) propose. A closer examination of the fund dimension reveals that funds that have a high dividend income or delivered high dividend payments to their investors reduce their ownership shares substantially more than funds without any dividend strategy. The differential response amounts to approximately 50% to 55% relative to funds without a dividend strategy. Importantly, funds that exhibit both characteristics, i.e., funds that pass dividends through to their investors, do not react more strongly than other dividend-oriented funds, providing little support for a pass-through mechanism. This conclusion is further supported by an alternative identification strategy that shifts the treatment dimension from banks to funds within the Eurozone and Swiss bank samples. Among Eurozone banks, active dividend funds decrease their ownership shares compared to a set of active funds without dividend strategy. In contrast, no such differential response is observed for Swiss banks. This even holds true when adding bank $\times$ month fixed effects and using the unmatched sample, alleviating concerns about unobserved differences in the bank dimension. This indicates that the observed ownership response is driven by changes in treated banks rather than by changes in Swiss banks.

Focusing on the bank dimension, there is no differential treatment effect between dividend smoothing and non-smoothing banks, which indicates that the alternative channel presented by Larkin, Leary and Michaely (2017) that mutual funds demand in particular smooth dividends is not supported by the data. Furthermore, the ownership reduction is not explained by differences in banks' default risk. Consequently, it is unlikely that the driving force is a rise in investors' expectations of higher bank defaults, because the intervention could have been seen as a negative signal about banks' stability. Differences in the Price/Book values can also not explain funds' ownership reductions. This suggests that banks subject to the free cash flow problem in the spirit of Jensen (1986) were not more affected than others.

The examination of dividend characteristics and fund ownership shares prior to the ban reveals a consistent and significant relation between fund ownership and banks' dividend yield. Specifically, a one percentage point increase in the dividend yield is related to an increase in fund ownership shares by 10%. This highlights that funds seem to be interested in dividends per se rather than a

smooth cash flow stream provided by dividend-smoothing banks.

Lastly, I investigate whether banks with greater ex ante exposure to fund investors experience different stock-market reactions around the policy announcements. I find a negative relationship between the percentage of fund owners per bank and the banks' cumulative abnormal returns (CAR) on the announcement of the policy. Affected banks with a higher fund ownership experienced a 7.5 percentage point lower CAR. This negative effect persisted during the announcement of limited dividend payouts and the abrogation of the policy, highlighting the long-lasting effects on the equity valuation of banks.

This study contributes to the literature on dividends and institutional investors. In their survey of firm executives Brav et al. (2005) find that corporate managers think households are the ones who care about dividends. This is in line with the demand-side approach to asset pricing of Kojien and Yogo (2019), where they verify that households have the strongest demand for dividend yields. Kojien, Richmond and Yogo (2024) further show that the dividend demand of institutional investors except for the private banking group is at best negative. I focus on institutional investors of banks and find that among those, funds are the investors who are interested in dividends, relying on an exogenous event instead of an instrumental variable estimation. Furthermore, I find that mutual funds' dividend demand is rather driven by the dividend level than its stability, as there is no evidence that dividend smoothing is impacting funds' ownership shares. This contrasts with the results of Larkin, Leary and Michaely (2017) for non-financial corporations. However, it is in line with the findings of Harris, Hartzmark and Solomon (2015), who analyze an extreme case of mutual funds dividend demand by excessive trading to capture dividends.

Additionally, another related strand of the literature is that of banks' payout policies. Abreu and Gulamhussen (2013) find that dividend payments for United States banks are in line with reducing the agency conflict of the free cash flow between shareholders and management (Jensen and Meckling, 1976; Jensen, 1986). Koussis and Makrominas (2019) analyze the dividend smoothing of European and US banks using the Lintner (1956) approach. They find that the agency conflict theory can explain dividend smoothing, but also that banks appear to smooth dividends less than non-financial firms. Building on this literature, I find that banks are also engaged in dividend smoothing, but to a lesser extent. Cziraki, Laux and Lóránth (2024) analyze the dividend policy of U.S. banks during the financial crisis and document that institutional investors did not push

banks to increase their dividend payments during the crisis. While they look at institutional investors' impact on banks' behavior during the financial crisis, I investigate how dividends and their restrictions affect institutional investors' behavior.

This paper also contributes to the growing literature examining the impact of sector-wide dividend restrictions during the COVID-19 pandemic (Hardy (2021); Martinez-Miera and Vegas Sánchez (2021); Dautović, Gambacorta and Reghezza (2023); Andreeva et al. (2023); Fringuellotti and Kroen (2024); Vadasz (2022); Marsh (2023); Federico, Modena and Regis (2025); Cáceres García and Lamas (2023); and Sanders, Simoens and Vander Vennet (2024)). Previous work focuses on the interaction of banks' risk-taking behavior, lending, equity valuation, and recapitalizations with dividend restrictions. This paper complements this literature by looking into institutional investors' reactions. Whereas Hardy (2021) gives a comprehensive overview of the reactions of policymakers around the world, Fringuellotti and Kroen (2024) find evidence for a reversal in risk shifting between equity holders and debt holders for US banks after the introduction of mainly share repurchase related payout restrictions. The interaction of the regulator, banks, and their investors after the regulator started to intrude into the payout policy of banks is modeled by Vadasz (2022) and Federico, Modena and Regis (2025). Vadasz (2022) shows that such discretionary ex-post interventions could even reduce the positive effects of dividend smoothing on risk management and bank value. Federico, Modena and Regis (2025) show that such policies could lead to higher dividend payments in good states to compensate for potential interventions, reducing capital buffers and recapitalization incentives. Complementary to these findings, I document that the temporary dividend restrictions have a long-term effect on the ownership structure of the banks. The lending channel of this policy is analyzed by Martinez-Miera and Vegas Sánchez (2021), Dautović, Gambacorta and Reghezza (2023) and Sanders, Simoens and Vander Vennet (2024). While Martinez-Miera and Vegas Sánchez (2021) focus on Spanish banks and use as identification the separation of dividend payers and non-payers during the restriction, Dautović, Gambacorta and Reghezza (2023) use supervisory data on distribution plans of significant institutions in the Eurozone to identify the effect. Both studies find a positive effect on lending to non-financial corporations. Sanders, Simoens and Vander Vennet (2024) also document an increase in syndicated lending and a negative stock price responses of European banks. However, the negative stock price reaction does not affect their lending results. Andreeva et al. (2023) analyze the impact of this regulation on bank

equity valuations also using the supervisory data on distribution plans, as Dautović, Gambacorta and Reghezza (2023) do. They find a negative impact on banks’ equity valuations, which is mostly driven by the uncertainty of future distributions and by banks that had planned to pay out dividends but could not live up to investors’ demanded returns. For the US setting, Marsh (2023) finds that larger banks faced negative stock price reactions to the announcement of dividend restrictions and spillover effects to unaffected banks. Lastly, in a concurrent study Cáceres García and Lamas (2023) analyze the reaction of income funds to the dividend restriction. While only focusing on the complete exit of the stock using Lipper data, they find that income funds temporarily exit affected banks’ stocks. Moreover, they find a negative stock price reaction at the announcement of the restrictions for banks with a higher exposure to income funds. In contrast, I focus on a different dimension affected by payout restrictions, namely the broader institutional ownership structure of banks. I find that among institutional investors funds reduce their ownership shares over the longer term, capturing not only exits but also reductions in ownership stakes. Moreover, I not only document that banks with higher ex ante fund ownership experience more negative announcement-window returns around the initial restriction announcement, but also at later selected payout-policy announcements, consistent with fund-investor exposure shaping the valuation effects of the policy.

2 Data

I draw on several data sources, with the main building block of the dataset being the monthly ownership data obtained from FactSet’s Ownership database. From this source, I retrieve total ownership shares for institutional investors, funds, and insiders/stakeholders, as well as investor-level ownership data for these three groups. FactSet’s ownership data are based on quarterly 13F filings and the monthly sum-of-funds data. 13F filings are required for institutional investors managing more than \$100 million in U.S. assets and are filed on a quarterly basis. Accordingly, in the empirical analysis in Section 3.2, I rely only on the sum-of-funds data when using the monthly frequency of the dataset. This separation prevents the monthly results from being downward biased due to lower time variation, as FactSet carries forward values from 13F filings within each quarter if no changes are reported. Furthermore, this approach alleviates issues related to selection bias in the owner dimension due to reporting requirements at the cost of lower ownership coverage

(see e.g. Steuer (2024), who compares the different reporting requirements in the US and EU and the impact on FactSet’s reported ownership)⁵. This limitation is also not too restrictive since *Table A2* shows that 85% of the observations are not from 13F filers and even for the group of institutional investors, which exhibits a larger share of 13F filers, the data still cover more than half of the observations in this group. Furthermore, *Figure A1* in Appendix A1 shows that the FactSet data is very comparable to the SHS dataset of the European Central Bank (ECB) that covers all investments of Eurozone investors in Eurozone banks’ listed shares. This also holds for the subset of the non-13F filing source.

Furthermore, to better extract funds’ ownership shares from other institutional investors’ ownership shares, I apply further data aggregations and transformations on the FactSet ownership data which are explained in detail in Appendix A1.

I also retrieve funds’ net flows, returns, size, and income return rankings from Morningstar. The data are aggregated to the fund level by taking the value-weighted average across the fund’s share classes. Missing values in the income return ranking are replaced with the value of the oldest share class. More details on the Morningstar data matching and variable creation can be found in Appendix Appendix A2

Balance sheet data for European banks are obtained from S&P Capital IQ, where I rely on the SNL Financials dataset. Market data and the data to calculate the dividend smoothing measures are taken from FactSet. Dividend expectations are taken from I/B/E/S. For the stock market reaction analysis, these data are complemented with the daily Fama/French European 3 Factors and 5 Factors available on Kenneth French’s website⁶. For the analysis using yearly ownership data in Section 3.3, I rely on the annual averages of aggregate fund ownership reported by FactSet.

I combine the datasets using ISINs and complement missing matches manually using Legal Entity Identifiers (LEI) and bank names. Significant institutions are identified from the March 2020 Single Supervisory Mechanism (SSM) lists⁷.

In total, the sample contains 66 European and Swiss banks of which four European banks already paid dividends and 12 European banks canceled their dividends before the restrictions

5. In the empirical section 3.2 the identification relies on individual owner-level data and not the total amount of all owners, thus, not having full coverage of all investors is not an issue as long as these investors are not systematically different. This would be the case by only relying on 13F filings for European stocks.

6. See https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

7. See <https://www.bankingsupervision.europa.eu/press/publications/html/index.en.html?skey=list>.

leading to 50 sample banks for the empirical analysis. In the investor dimension, the sample covers 5467 fund investors and 1831 institutional investors as of February 2020.

I limit price-to-book values to 20 following Larkin, Leary and Michaely (2017) and omit in the pre-intervention period analysis in section 3.3 observations with an aggregate ownership above 110%. I assume that aggregate ownership between 100% and 110% is due to measurement errors and truncate them to 100%. All monetary variables are measured in EUR and the later estimated smoothing measures as defined in Appendix A3 are winsorized at the 2.5% level.

Table 1 Panel A shows the descriptive statistics for the pre-intervention period for the sample of banks used in this study. Institutional investors hold on average 20% of bank equity, although ownership reaches up to 65% for some banks. Mutual funds account for roughly 17% on average, with ownership shares as high as 48%, indicating that funds constitute a major shareholder group for many banks. For the insider category, which also includes the government, we can see that there are banks that are almost completely held by this group. The dividend dimension of the data reveals a substantial dividend yield over the time period of on average 3.1%. Dividend smoothing is measured by the Speed of Adjustment (SOA) and Relative Volatility (RelVol), defined in Appendix A3. The smoothness of the dividend is measured by the variables Speed of Adjustment (SOA) and Relative Volatility (RelVol), which are defined in the Appendix A3. Lower values of both measures indicate stronger dividend smoothing. SOA captures how quickly dividends adjust towards their target payout, whereas RelVol measures the volatility of dividends relative to earnings. The average speed of adjustment of the sample banks is around 0.5. This value is much larger than the one estimated by Leary and Michaely (2011) for non-financial firms and Koussis and Makrominas (2019),, who report an average of roughly 0.3 for banks. This likely reflects both the shorter sample period and the inclusion of crisis years in the estimation. Consistent with this evidence, the average RelVol exceeds one, indicating that dividend payments are, on average, more volatile than their earnings based target over the sample period.

3 How Do Institutional Investors React to Dividend Restrictions?

To study the impact of dividend restrictions on institutional investors of banks' stocks, I rely on a quasi-natural experiment to identify the causal mechanism in a difference-in-differences setup. In

particular, I use the dividend restrictions implemented by the SSM in 2020 for the significant banks in the Eurozone as an exogenous shock to the payout policy (treated group), where Swiss banks are the control group. I restrict the period under study from 2018 to 2021 and use monthly data, to estimate the effect more precisely. The descriptive statistics for the monthly dataset of mutual funds and institutional investors excluding mutual funds are presented in *Table 2*.

The SSM dividend restrictions are a useful natural experiment for analyzing the implications of payout restrictions on the shareholder base. First, European banks conducted more dividend-smoothing compared to their US counterparts in the past (Koussis and Makrominas, 2019). This makes European banks particularly sensitive to dividend restriction policies because they provide a smooth cash flow to their investors. Second, in the Eurozone, payout restrictions had an impact on both dividends and repurchases, thereby preventing any kind of payout to the shareholders. This equal treatment among payout methods is important to rule out any shifts to alternative payout methods during that period, which would still yield a smooth income stream for investors. Hardy (2021) shows that this equal treatment was not the case for other jurisdictions, because for example, the dividend payments of US banks were largely unaffected, in contrast to repurchases, which decreased a lot. Third, the Eurozone was among the first regions to implement dividend restrictions, which limits the possibility of market participants anticipating the measure as the COVID-19 pandemic spread out and helps to estimate the effect more precisely.

3.1 Restricting Dividends in the Eurozone

3.1.1 Institutional Setup

On the 27th of March 2020, as a response to the threats to the banking sector caused by the COVID-19 pandemic, the ECB released a recommendation that stated that SIs⁸ should refrain from dividend payments and share repurchases. This measure was introduced at the consolidated group level of the significant institutions and was at first set to be in place until the 1st of October 2020 (ECB/2020/19). The justification for this recommendation was to prevent banks from distributing

8. A financial institution is classified as significant and then subject to the direct supervision of the SSM by the following criteria: (1) Total assets exceed € 30 bn., (2) is important for the country's economy or the whole EU, (3) total assets exceed € 5 bn. and cross border assets/liabilities to its total assets exceeds 20%, (4) requested or received financial aid from the European Stability Mechanism or the European Financial Stability Facility. See: <https://www.bankingsupervision.europa.eu/banking/list/criteria/html/index.en.html>

freed-up capital to their shareholders from the reduction in the buffer requirements, which was announced on the 12th of March in a press release⁹, and to make banks, in general, more resilient to the crisis by retaining capital. While at first aimed at SIs, many national supervisory authorities implemented the recommendation shortly afterward for less significant institutions (Beck et al., 2020).

Due to the ongoing COVID-19 pandemic in Europe, the ESRB also issued a recommendation to limit payouts for financial institutions until the end of the year 2020 at their meeting on the 27th of May (ESRB/2020/7)¹⁰. The recommendation of the ESRB was stricter than that of the ECB because it not only banned payouts until the end of the year, but also implemented restrictions on variable remuneration. The ECB issued, as a response, another recommendation on the 27th of July 2020, which extended the measures implemented in March until the 1st of January 2021 (ECB/2020/35). Furthermore, the variable remuneration restrictions were then also passed on to the banks supervised directly by the ECB via a letter¹¹.

On the 15th of December, the ECB again extended the recommendation, because the macroeconomic situation was improving but threats to the banking sector still remained due to the postponed impact on banks' balance sheets (ECB/2020/62). In contrast to previous notices, this time the supervisor acknowledged the improved economic situation and allowed again dividend payouts and repurchases, but these were limited to the lower of (i) 15% of cumulative profits for 2019 and 2020 or (ii) an amount corresponding to 20 basis points of their CET1 ratio at end-2020 (ECB/2020/62). This recommendation was valid until the end of September 2021. The official end of the restrictions was then announced on the 23rd of July with the recommendation ECB/2021/31, which verified that all dividend restrictions in place were to be lifted after the 30th of September 2021.

In contrast to this interference in the payout policy of SIs, the Swiss banking supervisor, the Swiss Financial Market Supervisory Authority (FINMA), took a different approach. On the 19th of March FINMA mentioned for the first time that financial institutions should follow a prudent payout policy¹². This comment was picked up on the 25th of March, when the warnings of non-

9. See: <https://www.bankingsupervision.europa.eu/press/pr/date/2020/html/ssm.pr200312~43351ac3ac.en.html>

10. Note that the public was informed later at the 8th of June in a press release of the ESRB: <https://www.esrb.europa.eu/news/pr/date/2020/html/esrb.pr200608~c9d71f035a.en.html>

11. See ECB press release of 28th July 2020: https://www.bankingsupervision.europa.eu/press/pr/date/2020/html/ssm.pr200728_1~42a74a0b86.en.html

12. See FINMA press release of 19th March 2020: <https://www.finma.ch/en/news/2020/03/20200319-mm-corona/>

prudent payout policies were intensified and it was recognized that Swiss banks had suspended their share repurchase program¹³. Finally, by the end of March 2020, FINMA issued the Guidance 02/2020, which again warned of non-prudent payout policies and summarized the capital relief measures that applied to banks' capital requirements. In contrast to banks supervised by the SSM, FINMA explicitly stated that if banks make profit distributions, this would lead to a reduction in the leverage ratio relief measures by the equivalent amount. Therefore, dividend distributions were not ruled out in Switzerland.

3.1.2 Banks' Dividend Responses and Dividend Expectations

These different approaches by the supervisory authorities in the SSM and in Switzerland make the two jurisdictions candidates for a difference-in-differences setup. First, it is important to note that both regions implemented their measures around the same time, i.e., end of March 2020. Furthermore, both implemented relief measures for capital requirements, which affect the stability and profitability of the institutions due to continued lending in times of crisis when capital ratios could fall. Additionally, both regions experienced similar impacts from the pandemic, as shown in *Figure A2* in the appendix, which illustrates that the number of new COVID-19 cases per million capita evolved in a similar manner. Lastly, a recent study by Hager and Nitschka (2023) shows that Swiss stock markets behave similarly to stock markets in the Eurozone in response to the ECB monetary policy surprises. This also alleviates the issue that ECB policies, like the Pandemic Emergency Purchase Programme (PEPP) drive the difference between the two groups.

To rule out voluntary dividend cuts of banks before the restriction was announced, I examined banks' ad-hoc announcements and excluded from the treated sample all banks with a dividend cut announcement before the restrictions were introduced. This ensures that the treated sample only contains banks that are exogenously affected by the dividend restrictions. Furthermore, I leave out banks that had already paid their dividends, since these banks pay quarterly or semi-annual dividends. This mitigates potential attenuation or bias in the estimates that could arise because the timing and status of any effect on these banks cannot be determined precisely¹⁴. This results

13. See FINMA press release of 25th March 2020: <https://www.finma.ch/en/news/2020/03/20200325-mm-garantiepaket/>

14. These banks are: Banco Bilbao Vizcaya Argentaria SA (ex-date 7th of April, announced 30th of January), Banco de Sabadell SA (ex-date 1st of April, announced 31st of January), Bankinter SA (ex-date 24th of March, announced 18th of February), CaixaBank SA (ex-date 9th of April, announced 30th of January). There were also banks from

in a sample of 50 banks, including 26 treated banks, and 24 control banks¹⁵.

Figure 1 Panel A shows the dividend yield as dividends over year-end market value and the total dividend yield, which includes repurchases for both groups. Dividend yields decreased significantly for treated Eurozone banks in the fiscal year 2019, while their Swiss counterparts did not significantly change their dividend yields. This is in line with the recommendation since dividends paid out in 2020 refer to the 2019 fiscal year, and the positive amount can be explained by interim dividend installments that banks had already paid out prior to the intervention, as well as shifted fiscal year-ends for Mediobanca. The rebound in the dividend yield in 2020 shows that banks were using the margin given by the SSM in December 2020 for the limited dividend payments, but this resulted in lower dividend yields compared to 2018. Repurchases followed dividends very closely for the treated banks, resulting in a parallel path of the dividend yield and the total dividend yield, whereas for the Swiss banks' average repurchases seem to have increased over the years although the confidence bands are quite wide.

Since dividends could also just fall because of lower distributable profits, *Figure 1* Panel B shows the payout ratio, as measured by dividend per share (DPS) over earnings per share (EPS), and the total payout ratio, which adds repurchases to DPS for the two groups. For treated banks, the analysis shows that for the 2019 dividends both payout ratios significantly fall, but in 2020 they already rebound close to their previous values. The set of control banks has a rather stable payout ratio although the average total payout ratio even increased from the fiscal year 2016 to 2020 and dropped in 2021. However, the values are not significantly different from their 2018 value. For the event year, payout ratios increased modestly for Swiss banks, which could be due to the pressure on their profits during the COVID-19 crisis. Overall, these results indicate that the recommendation was binding for the treated institutions, while Swiss banks' ratios were unaffected.

A first insight into the market reaction can be obtained from the dividend futures of European banks. I rely here on the Euro Stoxx Banks dividend future indices for different maturities to capture market expectations about Eurozone banks' dividends¹⁶. To isolate bank-specific changes

other countries that paid dividends in 2020, but these were non-listed banks and thus had missing data for relevant variables.

15. See appendix *Table A3* for a list of the banks in the sample

16. In February 2020, the constituents of the Euro Stoxx Banks index contained 24 Eurozone banks, of which only 2 were not group heads of SI groups: Finecobank S.p.A and Natixis. Therefore, the index is a good proxy for treated banks' dividend expectations.

from effects that affected the whole market, I divide the Euro Stoxx Banks dividend future indices by the respective Euro Stoxx 50 dividend future indices, where the latter captures market-wide effects on dividend expectations.

The evolution of the standardized bank dividend future series can be seen in *Figure 2* for the 2020 to 2023 contracts. Values are normalized to the values of each series one day before the dividend restriction announcement, i.e., the 26th of March 2020. The initial announcement mainly affected the 2020 dividend future, which lost more than half of its value within a few days and experienced a sharp increase in trading volume. By contrast, longer-maturity contracts initially increased and only declined after banks' first-quarter earnings releases, when banks typically update their payout policies, suggesting that markets initially interpreted the measure as a temporary postponement of dividend payments.

This interpretation changed following the ESRB recommendation in June, which appears to have signaled a prolongation of the restrictions, and the subsequent ECB extension in July. *Figure 2* shows that not only the 2020 contract but also the longer-maturity contracts declined after these announcements. The simultaneous decline across maturities suggests that markets revised down expected future dividend payments rather than simply shifting the 2020 payouts into subsequent years.

Towards the end of the year, several press interviews by SSM officials, marked by dotted lines in *Figure 2*, were accompanied by changes in dividend expectations. In particular, Andrea Enria's interview on 5 November signaled a "wait-and-see" approach¹⁷, while Financial Times reporting on 25 November suggested that dividend distributions would again be possible in 2021. Consistent with these signals, the 2021 dividend future increased ahead of the official announcement. However, after the ECB announced on 15 December that payouts would remain subject to strict quantitative limits, the 2021 contract declined again, indicating that markets had anticipated a more generous relaxation.

Panel B of *Figure 2* reinforces this interpretation by considering cumulative dividend expectations. The panel plots cumulative dividend expectations for Eurozone banks over the 2020–2022

17. Note that the spike in the traded volume on the 9th of November shortly afterward coincides with the success announcement of COVID-19 vaccines in trials, the final stage before marketing. See: Miller, Joe and Kuchler, Hannah and Mancini, Donato Paolo and Cookson, Clive. 2020. "Pfizer and BioNTech's Covid-19 vaccine found to be 90% effective" Financial Times, November 9. <https://www.ft.com/content/9bde4bff-acf0-4c2a-a0d0-5ed597186496>.

and 2021–2022 contract horizons relative to two benchmarks¹⁸. The series are reported relative to two benchmarks: EURO STOXX dividend futures, which control for aggregate market-wide dividend expectations and COVID-related risk, and IBES earnings expectations, which control for changes in expected bank earnings.¹⁹ Summing dividend expectations across maturities allows me to distinguish between a temporary postponement of dividends and a permanent revision of expected payouts. Following the restriction announcement, the 2020–2022 cumulative series decreases. However, the 2021–2022 cumulative series increases, consistent with markets expecting a temporary measure and a shift of dividends to future periods. Yet, the series does not increase sufficiently to be consistent with a complete shift of the 2020 dividends to subsequent years. At the same time, the 2020–2022 cumulative series remains well below its level prior to the COVID shock and the dividend restrictions. Thus, even if market participants expected a shift to later periods, this shift did not fully offset the reduction in cumulative expected dividend payments. Taken together, *Figure 2* provides little support for the view that the restrictions merely delayed dividend payments. Instead, markets appear to have revised down cumulative expected payouts over subsequent years.

3.2 Ownership Responses to Dividend Restrictions

Having established in the previous section that the restrictions for the treated group of banks were binding and that the control group was unaffected, I now turn to the main analysis of the paper, where I investigate the investor-level behavior in response to the dividend restrictions using an event study approach. One caveat of this analysis is that there could be concurrent events that affect investor behavior after the announcement (e.g. PEPP announcement). To overcome this issue and to isolate the treatment effect on the investor level, I rely on a within-investor identification strategy. Given the granular investor-level data, I include interacted fixed effects on the investor and time level. These fixed effects control for any time-varying effect per investor, i.e., investor demand side factors, and isolate effects within an investor. This approach is only valid if there are enough investors who invest in treated and control banks. The variable Ratio Treated-Control in

18. After a dividend future contract expires, its last observed value is carried forward when constructing the cumulative series. The same procedure is applied to the EPS expectations.

19. As IBES earnings forecasts refer to fiscal years, whereas dividend futures correspond to calendar-year dividends, I convert earnings forecasts to the calendar-year basis by weighting each fiscal-year earnings forecast according to the fraction of the fiscal year covered by the respective dividend future contract.

Table A2 represents the share of treated banks to all banks in the data an investor is invested in. Data on this variable is missing if an investor is not invested in any of the sample banks. This number is not zero or one for the 25th percentile to the 75th percentile for all groups except for the insider/stakeholder group, leaving enough heterogeneity in the data for the analysis. I also drop exchange traded funds (ETF) and funds with the FactSet holding style "Index" for this analysis to prevent a mechanical adjustment in the ownership due to benchmark index constituent changes²⁰. Lastly, I omit in the institutions' group the Norwegian Government Pension Fund Global, because this fund has large ownership shares that change mostly at the end of the year. This leads to block-like patterns in the monthly event-study standard errors as evident in *Figure A4*, where the fund is included, yet these results are consistent with the main results without the fund, presented in *Figure 3*.

One possible issue in the proposed analysis could be that the sample of Swiss banks does not represent a good control group for significant banks in the Eurozone. On the one hand, the significance status of Eurozone banks is mainly determined by size, whereas in the control group a large share of small banks are also included. On the other hand, Swiss banks could be more resilient by having higher capital ratios and Price/Book values. Although FINMA mentions in its decision that the Swiss banking sector is resilient, the ECB also stated that European banks are resilient but nevertheless implemented the restrictions. Ultimately, it is not clear what measures influenced the supervisors' final decisions. It might be that the ECB took measures to limit banks' dividends because larger banks have a higher probability to pay dividends (in particular during crises) as Abreu and Gulamhussen (2013) show, or because they have lower franchise values and thus are more exposed to the free cash flow problem. This issue might not be overcome by just controlling for size in the regressions via bank-fixed effects. Furthermore, there could also be unobserved confounders that affect the treatment status and the reaction of investors. For example, Swiss banks could have different business models despite operating in similar markets²¹.

To alleviate these issues, I conducted a propensity score matching to make the treatment group and control group of banks more comparable. I use k-nearest neighbor matching with a maximum of two control banks per treated bank and set the threshold for the propensity score to 0.1. I match

20. *Figure A4* in the appendix indicates that ETFs were also not affected by the restriction.

21. Note that for 15 out of 25 significant institutions in the sample that have segmented net income available, on average 68% of their net income is domestic, indicating that these banks mainly operate in Europe.

on the already mentioned size, Price/Book ratio (a proxy for the franchise value), and the Tier1 capital ratio values one year before the dividend restrictions, i.e., in 2019. This initial matching step results in 27 comparable banks. To overcome common support issues, I trim the matched sample by excluding all control banks that have a lower propensity score than the lowest propensity score of the treated banks. This effectively removes only one control bank, resulting in 26 matched banks. The sample means of the matched treated and control groups can be found in *Table 1* Panel B. After trimming, the treated and control group averages are quite similar for the matched variables size, Price/Book ratio, and the Tier1 capital ratio. Additionally, the averages of the smoothing proxies RelVol and SOA, the proxy for the banks' business model Loans/Deposits, the percentage of institutional investors holding the bank's stock, and the return on assets in 2019 are very similar. The t-tests of the means for these groups are insignificant and indicate a good match. Lastly, to see if the two groups are similar in forward-looking measures, the last four rows of *Table 1* Panel B contain the weighted average cost of debt in the years 2019 and 2020 and the I/B/E/S expectations for the 2020 dividend yield and payout ratio.²² Again, except for the expected dividend yield that is significant only at the 10% level, the averages of the two groups are statistically not different from each other and thus indicate that funding costs as well as payout expectations for the treated banks were similar before and during the restriction period.

This matched sample can now be used to look into the overall impact of the regulations, by using a classical difference-in-differences approach, because all banks are treated at the same time. The treatment timing indicator is split in this setting into three different bins, each representing a different period of the regulation: March to November 2020 (restriction), December 2020 to June 2021 (relaxation), and July 2021 onwards (expiration). Since ownership shares are right-skewed and the investor-level data also contains zeros, I follow the suggestions of Cohn, Liu and Wardlaw (2022) and Silva and Tenreiro (2011), and apply instead of a $\text{Log}(1 + y)$ transformation for the dependent variable, a Poisson pseudo-maximum likelihood regressions to estimate the effect on ownership percentages. This approach allows me to also include multi-way fixed effects but circumvents the incidental parameter bias.

22. I use the weighted average cost of debt provided by FactSet, as Credit Default Swaps (CDS) spreads are not available for many control banks.

The model is as follows:

$$\begin{aligned}
Ownershare_{b,i,t} = & \exp\left(\alpha + \delta_1 Treated_b \times \mathbb{1}(Restriction)_t \right. \\
& + \delta_2 Treated_b \times \mathbb{1}(Relaxation)_t \\
& + \delta_3 Treated_b \times \mathbb{1}(Expiration)_t \\
& \left. + \phi' X_{b,t} + \gamma_{t,i} + \gamma_b\right) + \varepsilon_{b,i,t}
\end{aligned} \tag{1}$$

where $Ownershare_{b,i,t}$ is the % $Ownershare_{b,i,t}$ for investor i at month t in bank b . $\mathbb{1}(Restriction)_t$ is an indicator variable for the period March 2020 to November 2020, $\mathbb{1}(Relaxation)_t$ for the period December 2020 to June 2021, and $\mathbb{1}(Expiration)_t$ for the period on and after July 2021. $Treated_b$ is the treatment group indicator for each bank b . $X_{b,t}$ are bank-level control variables, i.e., the change in the shares outstanding to eliminate changes induced by the company by issuance or repurchases (for the control group), and the exchange rate to EUR. In an additional specification I also add the lagged stock return over the month of a bank, the lagged volatility of the bank's daily stock returns over the past 21 business days to control for changes due to the performance of the stocks. $\gamma_{t,i}$ is the *investor* $\times$ *month* fixed effect to control for investor demand-side effects and γ_b are the bank fixed effects. The coefficients of interest are δ_1 , δ_2 , and δ_3 , which measure the restriction impact, relaxation impact, and the impact after the expiration of the restriction, respectively.

Table 3 shows the baseline results for funds. Without controlling for investor-specific time-varying demand in Column (1), the estimated treatment effects are small and statistically insignificant. Once *investor* $\times$ *month* fixed effects are introduced, the ownership response becomes negative and statistically significant at the 1% level throughout all three policy periods, indicating a persistent reduction in fund ownership following the dividend restrictions. The effect also becomes economically larger over time from -0.157 to -0.388. This represents a decrease in the average fund ownership in the restriction period of $(\exp(-0.157) - 1) * 100 = -14.5$ % per investor. Adding additional controls for risk and return, i.e., the lagged monthly return and the stock volatility, in column (3) does not change the coefficients' magnitude nor the precision of the estimates, as the standard errors are quite similar.

As an additional robustness check, I conduct randomization inference, which exploits the known

random assignment of the treatment across banks and provides design-based inference that does not rely on large-sample approximations. Given the non-linear nature of the estimation I also conduct a randomization inference for the last two columns, where I permute the treatment assignment across banks without replacement. The p-values based on 2,000 random permutations under the one-sided alternative hypothesis that the treatment effect is negative are at the bottom of *Table 3*. For both models the coefficients for the relaxation and expiration period are significant at least at the 10% level. The insignificant estimate during the restriction period does not necessarily imply the absence of a treatment effect, but may instead reflect its gradual evolution over time. This aspect is examined in more detail in the event-study analysis below.

While funds seem to be immediately affected by the restriction, *Table 4* shows the results for institutions other than funds. Again, the same specifications are applied as for funds, meaning column (1) shows the results without the investor $\times$ month fixed effects, while (2) and (3) use investor $\times$ month and bank fixed effects. In contrast to the fund results, none of the treatment effects is statistically significant across the different specifications, suggesting that the ownership response is specific to funds rather than other institutional investors. This is in line with the findings of Larkin, Leary and Michaely (2017) and Harris, Hartzmark and Solomon (2015), which find that funds are investors that demand dividends. However, the estimates are also less precise because the standard errors of the coefficients are much larger than the ones in the specification for funds. This is not only due to the smaller sample size but could also be attributed to the higher overall variation in ownership shares for institutional investors, as evidenced in *Table 2*.

As for the fund sample, I also conduct randomization inference for the institutional investor sample. The p-values from this draws under the alternative hypothesis that the estimates are negative are at the bottom of *Table 4*. For both specifications, none of the estimated treatment effects is statistically significant at the 10% level, providing additional support for the absence of a systematic ownership response among non-fund institutional investors.

Due to the dynamic nature of the treatment effect of the restriction, I also estimate an event study Poisson pseudo-maximum likelihood regression to see the evolution of the treatment effect. In particular, I follow Freyaldenhoven et al. (2026), where the setting under study has the advantage of no staggered implementation. Let τ be the month of the implementation of the restriction, i.e.,

March 2020, then the event study model is as follows:

$$\begin{aligned}
Ownershare_{b,i,t} = & \exp\left(\alpha + \beta_{-10}\mathbb{1}(t \leq \tau - 10) + \sum_{k=-9}^{18} \beta_k \mathbb{1}(t = \tau + k) + \beta_{19}\mathbb{1}(t \geq \tau + 19)\right. \\
& + \delta_{-10}\mathbb{1}(t \leq \tau - 10) \times Treated_b \\
& + \sum_{k=-9}^{18} \delta_k \mathbb{1}(t = \tau + k) \times Treated_b \\
& + \delta_{19}\mathbb{1}(t \geq \tau + 19) \times Treated_b \\
& \left. + \phi' X_{b,t} + \gamma_{t,i} + \gamma_b\right) + \varepsilon_{i,t,b}
\end{aligned} \tag{2}$$

where $Ownershare_{b,i,t}$ is the % $Ownershare_{b,i,t}$ for investor i at month t in bank b . $Treated_b$ is the treatment group indicator for each bank b . $X_{b,t}$ are bank-level control variables, i.e., the change in the shares outstanding, to eliminate changes induced by the company by issuance or repurchases (for the control group), and the exchange rate to EUR. $\gamma_{t,i}$ is the *investor* $\times$ *month* fixed effect to control for investor demand-side effects and γ_b are the bank fixed effects. The coefficients of interest are in the array of $\{\delta_k\}$, where δ_{-10} captures all time periods on and before May 2019 and δ_{-9} to δ_{-1} would capture pre-event trends and help to support the parallel trend assumption. The impact of the dividend restrictions is captured by the coefficients δ_0 to δ_{16} because they span the time horizon from March 2020 to July 2021, when it was announced that the dividend restrictions would not be extended. To overcome the multicollinearity issues associated with the indicator variables, I follow the suggestion of Freyaldenhoven et al. (2026) and normalize the event study plots' coefficients to the February 2020 value, given that the impact of COVID-19 on financial markets in Europe began in February. Furthermore, I add sup-t confidence bands proposed by Montiel Olea and Plagborg-Møller (2019) for the event study plots to draw conclusions from the joint significance of the event path rather than its individual significance at certain points in time.

The event study plots of the regression model described in equation (2) can be found in *Figure 3*, Panels A and B for each sub-sample, while the regression tables are *Table A5* and *A6* in the Appendix A4. For funds, in Panel A of *Figure 3*, the pre-announcement coefficients are economically small and statistically insignificant, supporting the parallel trends assumption. Following the announcement of the dividend restrictions, ownership shares decline gradually, becoming sta-

tistically significant at the 5% level from June 2020 onward and jointly significant from August onward. By November 2020, the average ownership share per fund is around -19.5% lower than in the control group and continues to decline to approximately -28% by June 2021, indicating a persistent reduction in fund ownership. In contrast to the effect evident in *Figure 3* Panel A, the results for institutional investors other than funds in Panel B show no significant effect using the joint significance bands, although some individual months show negative significant effects using pointwise confidence intervals.

As shown in *Table 3*, adding the stock return of the previous month and the daily return volatility does not alter the results.

The previous estimates showed the combined effect on the intensive and extensive margin, i.e., changes in existing ownership shares as well as entry and exit decisions. While this shows the average treatment effect on fund ownership, the extensive margin is of independent interest because mutual fund ownership has been linked to stronger governance (Appel, Gormley and Keim, 2016) and lower bank risk-taking (Pathan et al., 2021). Consequently, a complete exit of fund investors may have broader implications for banks than a mere reduction in ownership stakes. *Figure 3*, Panel C, shows the linear probability model estimation of equation (2), where the dependent variable equals one if investor i is invested in the bank b at time t and utilizes the same control variables and fixed effects on the right hand side as (2). The probability of holding a treated bank begins to decline in April 2020 and becomes statistically significant from June onwards, coinciding with the ESRB recommendation that first signaled an extension of the restrictions. By November 2020 the estimates indicate an approximately 10% lower holding probability. The negative effect remains jointly significant until February 2021. Although the estimates partially recover following the restart of dividend payments in spring 2021, they remain negative throughout the sample period. Panel D shows the results for institutional investors other than funds. Here we can also see a significant decrease in the pseudo probability, although the decrease is more abrupt, smaller, and partly reverts in March 2021. The effect is not as pronounced as for fund investors, since for example the estimate in November 2020 only amounts to a reduction of 8%.

Appendix A4 *Figure A3* also shows the results for the unmatched sample. Here, the reaction of funds is lower, not persistent, and statistically insignificant as evident in *Figure A3*, Panel A. For institutions using the joint confidence bands, there is also no effect evident, although the parallel

trend assumption cannot be supported here due to the non-parallel pre-event trends. However, regarding the exit decision, the effects seem to be similar. Note that the unmatched sample contains substantially greater heterogeneity in bank characteristics because it includes all Swiss and Eurozone banks irrespective of comparability. Consequently, treatment and control banks are less comparable, resulting in less precise estimates and leaving the baseline difference-in-differences estimates more exposed to time-varying bank-level heterogeneity. This motivates the matched-sample design used throughout the main analysis. The role of the matched sample is revisited in the mechanism analysis, where the richer fixed-effects structure allows the estimates to be obtained on the full sample of banks while controlling for bank level confounders.

3.2.1 Dividend Funds

The analysis thus far has shown that funds reacted to dividend restrictions. Yet, the motivation behind this behavior remains unclear. In line with the findings of Harris, Hartzmark and Solomon (2015), one possibility is that funds holding bank stocks have their own dividend-distribution objectives and therefore prefer dividend-paying bank stocks in order to pass these dividends on to their own investors. This channel would be consistent with funds that both receive high dividend income and make high dividend payments to their investors. Alternatively, funds may either pursue a simple dividend-income strategy by investing in dividend-paying stocks without passing on that dividend income as dividends, or maintain a high dividend-distribution strategy even when fund distributions are not primarily financed by dividend income.

To investigate which mechanism is at work, I use the subset of the dataset that can be matched to fund-level data from Morningstar. Instead of relying on fund names or investment classifications, I directly measure both (i) the dividend distributions made by the fund to its investors and (ii) the dividend income received from the underlying equity portfolio. These measures provide a considerably more direct characterization of dividend-oriented investment strategies. In particular, (i) is measured by the income return percentile ranking of funds within their Morningstar category in 2019, where the return is based on the fund’s dividend distributions that are reinvested into the fund until the year-end and are then ranked. Funds with a ranking of 1 are the highest dividend payers in the Morningstar universe, i.e., these funds distributed more dividends in 2019 than their peers.²³

23. The correlation of the 2019 income return measure with its 2018 value is around 0.7.

The latter (ii) is constructed from monthly fund holdings matched with Compustat Global and North America stock data to attribute the stock’s dividends to the funds’ holdings.²⁴ In particular, it sums the dividends received within 2019 for a fund and divides them by the fund’s net asset value of the year before. Appendix A2 show more details on the construction of the dividend income variable and also checks potential selection of the sample due to the availability of the Morningstar data. Table 2 shows that the median value of received dividends in relation to the fund’s assets is 3.322% and the median income return of the funds in 2019 is 2.085%. The latter results in a median income return ranking of 33, indicating that funds invested in the sample banks are in the upper third of the dividend payers.

To distinguish the channel through which funds react to dividend restrictions, I construct a categorical variable based on the interaction of two indicators. First, *HighPay_i* equals 1 for funds with an income return ranking of 33 or lower, identifying high dividend-paying funds. Second, *HighInc_i* equals 1 for funds with above-median dividend income received from portfolio holdings in 2019. This categorical variable thus spans the different combinations of high and low fund payouts and high and low dividend income received from holdings, allowing me to distinguish whether the response is driven by funds’ own payout policy, their portfolio dividend income, or both. Throughout, I restrict the sample to funds observed over the full estimation period, i.e., 2018 to 2021.

I then use these two dummy variables and their interaction in a triple difference setup where I interact the interaction terms of equation (1) with the additional indicator variables for funds with high dividend payments and funds with high dividend income and their interaction term:

24. This variable is based on funds that have at least 50% of their holdings invested in equities and of which at least 80% could be matched with compustat each month.

$$\begin{aligned}
\text{Ownership}_{b,i,t} = & \exp\left(\alpha + \sum_E \left(\psi_{1,E} \text{Treated}_b \times \mathbb{1}(E)_t\right.\right. \\
& + \left[\delta_{4,E} \mathbf{HighPay}_i + \delta_{5,E} \mathbf{HighInc}_i + \delta_{6,E} \mathbf{HighPay}_i \times \mathbf{HighInc}_i\right] \times \text{Treated}_b \times \mathbb{1}(E)_t \\
& + \left[\delta_1 \mathbf{HighPay}_i + \delta_2 \mathbf{HighInc}_i + \delta_3 \mathbf{HighPay}_i \times \mathbf{HighInc}_i\right] \times \text{Treated}_b \\
& \left. + X_{b,t} + \gamma_{t,i} + \gamma_b\right) + \varepsilon_{b,i,t}
\end{aligned} \tag{3}$$

where $E \in \{\text{Restriction, Relaxation, Expiration}\}$

where $\text{Ownership}_{b,i,t}$ is again $\% \text{Ownership}_{b,i,t}$ and HighPay_i is the indicator variable for high dividend paying funds, and HighInc_i is the indicator variable for funds with high dividend income from their holdings. Treated is one for banks that are subject to the restrictions and did not already pay out dividends in 2020 or cut their dividends before the announcement. $X_{b,t}$ are the control variables used in the previous specification. $\gamma_{t,i}$ are again the investor $\times$ month fixed effects and γ_b are the bank fixed effects.

The results are presented in *Table 5*, column (1). Across the different time horizons, we see that funds without a dividend-oriented strategy exhibit no meaningful response to the dividend restrictions, as only in the relaxation period is the treatment coefficient significant, and this also only at the 10% level. If anything, the point estimates are positive, indicating that these funds did not reduce their ownership. Turning to the differential response of high-dividend-paying funds, i.e., *HighPay*, relative to the baseline group, all interaction coefficients are negative and statistically significant at the 1% level. We can even see a further decrease in the point estimates over time from -0.803 in the restriction period to -1.092 in the expiration period, which corresponds to an additional ownership reduction of approximately 55% to 66% relative to the baseline group. Funds with high dividend income (*HighInc*) also reduce their ownership more strongly over time. The differential effects are statistically significant at the 5% level during the restriction and relaxation periods and at the 10% level during the expiration period. The interaction term provides the direct test of the pass-through hypothesis, as it measures whether funds that both receive high dividend income and distribute high dividends react more strongly than implied by the two individual dividend characteristics. Contrary to the pass-through hypothesis, the interaction coefficients are

positive and statistically significant at the 1% level, except for the expiration period, where they are significant at the 5% level. Hence, funds satisfying both dividend characteristics do not reduce their ownership more than funds satisfying only one of the two characteristics. Indeed, summing the relevant coefficients shows that their implied response is similar in magnitude to that of funds satisfying only one of the two dividend criteria. Thus, while dividend-oriented funds respond more strongly than non-dividend funds, there is little evidence that pass-through funds respond more strongly than other dividend-oriented funds.

One remaining concern is that the stronger reaction of dividend-oriented funds could still reflect differential developments across treated and control banks during the pandemic rather than differences in investors' dividend preferences. For example, dividend-oriented funds might simply have held banks that experienced larger deteriorations in profitability or capitalization. To address this concern, column (2) augments the specification with bank $\times$ month fixed effects. Consequently, identification comes exclusively from comparing dividend-oriented and non-dividend funds holding the same bank in the same month, thereby absorbing all contemporaneous bank-level shocks. The point estimates remain virtually unchanged relative to column (1), and their statistical significance is unaffected. This stability indicates that the differential response of dividend-oriented funds is unlikely to be driven by contemporaneous bank-level developments, providing further support for the interpretation that the results reflect differences in investors' dividend preferences.

Lastly, column (3) further relaxes the reliance on the matched sample by estimating the model on the full bank sample while continuing to absorb all contemporaneous bank-level heterogeneity through bank $\times$ month fixed effects. Comparing the estimates from column (3) to (2) we can see that the results are nearly the same in terms of direction and significance, albeit the magnitude is smaller. One difference appears for the expiration period where we see that the *HighPay* funds do not show a negative significant effect anymore. Their point estimate is around zero and insignificant. Moreover, the interaction term of the *HighPay* funds and the *HighInc* funds in this treatment period is also insignificant and also around zero. Thus, in the unmatched sample, the differential response of *HighPay* funds disappears during the expiration period. Consequently, the implied response of the pass-through funds remains similar to that of the *HighInc* funds. Summing the interaction coefficients yields an implied effect in the restriction period of -0.56 for pass-through funds, compared with -0.591 for *HighPay* funds and -0.688 for *HighInc* funds. However, at expi-

ration, this changes to -0.53 for pass-through funds, 0 for HighPay funds and -0.704 for HighInc funds.

Figure 4 visualizes the implied treatment effects underlying the interaction specification in *Table 5*. The figure displays the level estimates from the event-study specification for each fund category defined by the two indicator variables *HighInc* and *HighPay*. Starting with the gray series that represents funds without high dividend income or high dividend payments ($HighInc = 0$ and $HighPay = 0$), the pre-treatment estimates are tightly centered around zero and statistically insignificant, indicating no evidence of differential pre-trends. Following the announcement of the dividend restrictions, the point estimates remain close to zero and continue to be statistically insignificant, suggesting that these funds were largely unaffected by the policy.

For funds with only high dividend payments ($HighInc = 0$ and $HighPay = 1$) in green, the pre-treatment estimates' joint significance bands indicate no statistically significant pre-treatment dynamics. After the announcement of the dividend restrictions, the estimates become negative and exhibit a pronounced decline in June 2020, pushing the point estimates significantly below zero. Although the joint significance confidence bands become insignificant again from March 2021 onward, the conventional 95% confidence intervals remain below zero for most of the remaining sample period.

Turning to funds with only high dividend income ($HighInc = 1$ and $HighPay = 0$) in light blue, the pre-treatment estimates also provide no evidence of systematic pre-treatment dynamics. Following the announcement of the policy, the point estimates become negative and remain below zero throughout the post-treatment period. The conventional 95% confidence intervals indicate statistically significant effects for most post-treatment months, while the joint significance confidence bands are significant from September 2020 onward, except for the period between March and June 2021. Lastly, funds with both high dividend income and high dividend payments ($HighInc = 1$ and $HighPay = 1$) in navy also exhibit no evidence of pre-treatment trends, as the pre-treatment estimates are statistically insignificant. After the treatment, the point estimates decline steadily, indicating a gradual negative response to the dividend restrictions. Compared with funds that only have high dividend payments, the decline is more gradual and smaller in magnitude, but its evolution closely resembles that of the funds with only high dividend income.

Overall, the event-study estimates provide further support for the baseline results by visualizing

the implied treatment effects over time underlying the interaction specification in *Table 5*. Consistent with the interaction estimates, the three dividend-oriented fund categories exhibit ownership reductions of broadly similar magnitude throughout most of the post-treatment period. Interestingly, the plot reveals that funds with high dividend payments only display a somewhat stronger response between the ESRB recommendation in June 2020 and November 2020 before converging toward the other dividend oriented fund groups. Except for this episode, the confidence intervals overlap substantially throughout most of the post-treatment period. Together, these results provide little support for a pass-through mechanism under which funds passing dividend income through to their investors would be expected to respond more strongly than other dividend-oriented funds.

Finally, given the additional dimension through the dividend funds by the previously established mechanism it is also possible to split the treated and control sample of banks and run the test on each of those subsets, by changing treatment from the bank level to the fund level.

In particular, the dimensionality of the data allows me to use dividend funds as the treatment set and funds without dividend characteristics, i.e., low dividend income and low dividend paying funds, as control set. I apply this treatment specification on the set of Eurozone banks, and separately on the set of Swiss banks. If dividend funds were the ones reacting to the dividend restrictions, we would observe that those funds would decrease their ownership at Eurozone banks compared to funds without dividend characteristics. Likewise, dividend funds would not change their ownership share at control banks compared to non-dividend funds, showing that there is no reallocation from Eurozone banks to Swiss banks and that the estimated effect is not driven by funds perceiving Swiss banks as a safe haven in crisis times.

Therefore, I adapt equation 4 as follows:

$$\begin{aligned}
Ownership_{b,i,t} = & \exp\left(\alpha + \delta_{-10}\mathbb{1}(t \leq \tau - 10) \times \mathbf{Treated}_i \right. \\
& + \sum_{k=-9}^{18} \delta_k \mathbb{1}(t = \tau + k) \times \mathbf{Treated}_i \\
& + \delta_{19}\mathbb{1}(t \geq \tau + 19) \times \mathbf{Treated}_i \\
& \left. + \phi' \mathbf{X}_{i,t} + \gamma_{b,t} + \gamma_{s,t} + \gamma_i\right) + \varepsilon_{i,t,b}
\end{aligned} \tag{4}$$

where $Treated_i$ is an indicator variable equal to one for dividend funds, i.e., funds with either high

dividend income or high dividend payments, $X_{i,t}$ contain the fund time-varying control variables lagged monthly return, lagged daily return volatility over the last month, and lagged monthly net fund flows. $\gamma_{b,t}$ are the bank $\times$ month fixed effects, $\gamma_{s,t}$ are the fund strategy $\times$ month fixed effects, based on the Morningstar category the fund is classified, and γ_i are the fund fixed effects.

Figure 5 displays the δ coefficients of this regression for the different samples. Starting with the sample of matched Eurozone banks in Panel A, it is evident that dividend funds ownership shares follow the same pattern as the treatment effect estimated in Figure 3. However, the effect is not as strong, as the joint significance bands are insignificant at the 5% level, while the individual significance bands partly show significance at the 5% level. Note that the coefficients from May 2020 to September 2021 are all significant at the 10% significance level using the individual confidence intervals. Furthermore, the pre-period estimates are almost significantly positive under the joint significance bands, but a formal test for the 6 months before the treatment coefficients being jointly zero returns a χ^2 value of 8.616 with a p-value of 0.125, indicating insignificant pre-trends at conventional significance levels. Turning now to the Swiss banks in the matched sample in Panel B, one can see that the post-event estimates are positive and are all insignificant at the 5% level using the individual confidence bands as well as the joint significance bands. An interesting observation in this plot is the binned estimate for the periods until May 2019. It shows a larger negative effect compared to all the other estimates, but is still insignificant using both confidence bands.

One natural question is whether these patterns also hold for the unmatched sample. This can be seen in Panel C for the EU banks and in Panel D for the Swiss banks. Starting off with the Eurozone banks, a similar pattern as the one in Panel A can be seen; however, the estimates are estimated under higher uncertainty as their confidence bands are wider. Regarding the pre period estimates, the two months before the event are well aligned around zero, yet, the earlier estimates seem to be positively shifted upward and two of them are also significant using the joint significance bands. A formal test of the six months before the event coefficients being zero returns a χ^2 value of 10.586 with a p-value of 0.06, rejecting the joint significance only at the 10% level. Turning now to the Swiss banks' effects, it is evident in Panel D that those do not react to the dividend restrictions, as all estimates are insignificant using the individual as well as the joint significance bands. Also, the pre period estimates are centered around zero except the estimates before June 2019 are negative

and the binned one is also significant at the 5% level using the individual confidence band. While the unmatched Eurozone panel is somewhat noisier in the pre-period, the results still suggest that the differential response is not driven solely by the matched sample.

Taken together, the results remain qualitatively similar in the unmatched sample, alleviating concerns that the results are driven by the matched sample. Taken together, the bank $\times$ month fixed-effects specification and the within-Eurozone analysis substantially alleviate concerns that the estimated effects are driven by contemporaneous bank-level shocks or by systematic differences between Swiss and Eurozone banks. Instead, the evidence points to differential responses across investor types facing the same bank-level information, consistent with dividend-oriented funds reacting specifically to the supervisory dividend restrictions.

3.2.2 Are Some Banks More Affected Than Others?

Given the results in the previous section that dividend funds reacted to dividend restrictions by reducing their ownership share, I now take a closer look at whether some banks are more affected than others. Given the evidence documented by Larkin, Leary and Michaely (2017), it could be that banks that smooth their dividend stream are more affected compared to non-smoothers, because funds want to have a stable cash flow to pay out cash themselves. This would highlight the dividend smoothing channel of the investors' reaction, as dividend restrictions have a strong impact on the smoothness of the cash flow streams to investors. If instead, funds are only interested in the dividend yield, we would expect that banks with higher pre-intervention dividends experience a higher reduction in the ownership of funds. Furthermore, low capitalization or low Price/Book values could also explain the portfolio adjustment of funds if the underlying problem of the exit is instead the free cash flow problem between shareholders and management (Jensen and Meckling, 1976; Jensen, 1986). Lastly, if the dividend restrictions signal to investors that banks are not resilient, one would expect that banks with a lower distance to default or capitalization would experience a higher reduction in the ownership of fund investors.

To test which of the proposed mechanisms is at play, I again extend the model in equation (2) with an additional dummy variable $Char_b$ and interact this variable with the treatment dummy to obtain a triple difference model. $Char_b$ is an indicator variable equal to one for banks below the median value of the matched sample of either RelVol, SOA in the fiscal year 2018 or the analyst

dividend yield expectation for 2020, the below median value of the Price/Book, the Tier 1 capital ratio and the logarithm of the Z-score in 2019, respectively.

Table 6 shows the results for the average effect of $Char_b$ being the below-median value of either RelVol, SOA, the dividend yield expectations, Price/Book values, distance to default as measured by $\text{Log}(Z - \text{Score})$, or capitalization. Beginning with the dividend dimension and the speed of adjustment measure of smoothing, column (1) shows that there is no statistically significant effect for the triple interaction terms with SOA throughout the periods. Also, for the smoothing measure of RelVol in column (2) there is no statistically significant effect across all time periods. Taken together, I do not find evidence that dividend smoothing drives the results, since there is no statistically significant effect across the smoothing proxies. Regarding the dividend yield expectations, one would expect that banks with lower dividend yields should have a positive coefficient for the triple interaction term. Column (3) shows that for the restriction period, the point estimates are negative but statistically insignificant. This implies that there was no differential impact for banks with different dividend yield expectations.

If the agency problem between the shareholder and the management is the driver of the results, one would expect a negative coefficient on the low Price/Book values in column (4) of *Table 6*. Indeed, column (4) shows negative point coefficients, however, they are insignificant. *Figure A5*, Panel A also supports the parallel trend assumption and confirms that the ownership percentages were not changing. Thus, the free cash flow problem seems not to explain the results. Lastly, if investors are reacting only because the dividend restrictions are a negative signal about banks' resilience, one would expect a stronger reduction for banks with a low $\text{Log}(Z - \text{Score})$ in column (5) or capital ratio in column (6). Yet, the coefficients for the $\text{Log}(Z - \text{Score})$ in column (5) and for the low capital ratio in column (6) have positive signs during the restriction period and the relaxation period. For banks with a lower distance to default, this coefficient is even statistically significant at the 5 % level. This indicates that fund investors were not reacting to a negative signal about banks' resilience.

3.3 What Affected Fund Ownership Before The Intervention?

The previous findings suggest that fund investors of banks have a demand for dividends, but not necessarily smooth dividends, and thus reduce their ownership of banks' stocks after the announce-

ment of the dividend restrictions. One drawback of the previous analysis could be that there is not enough variation left to identify the effect of the dividend smoothing due to the small sample size of the matched sample in this analysis, because smoothing measures could not be calculated for all banks. To shed further light on the question of whether fund investors have a higher demand for dividend yield or for smooth dividends, I buttress the analysis by looking at the determinants of fund ownership of European banks before the dividend restriction period. If the decrease was driven by fund investors who want to receive a smooth cash flow stream, I would find a negative impact of the dividend smoothing proxies on fund ownership already before the restriction period. Conversely, if fund ownership is instead driven by the dividend yield, there would be a positive relation between the dividend yield and fund ownership even before the restriction period. To disentangle the two mechanisms, I use a similar approach as Larkin, Leary and Michaely (2017) and regress aggregate fund ownership at the bank level on dividend smoothing proxies, the dividend yield, and other control variables over the period from 2016 to 2019 using the following model:

$$FundShare_{b,t} = \exp\left(\beta_0 + \beta_1 Smooth_{b,t-1} + \beta_2 Div.Yield_{b,t-1} + \delta' X_{b,t-1} + \gamma_c + \varepsilon_{b,t}\right) \quad (5)$$

where $Smooth_{b,t-1}$ is either $SOA_{b,t-1}$ or $RelVol_{b,t-1}$ of bank b at year $t - 1$. $Div.Yield_{b,t-1}$ is the dividend yield of bank b at year $t - 1$. $FundShare_{b,t}$ is the ownership share of funds for each bank and year. I control in this setting for lagged investor controls contained in $X_{b,t-1}$. Namely, the Price/Book ratio, the stock return over each last fiscal year as a proxy for momentum, the daily stock price volatility over the last year for risk, the logarithm of total assets at the end of the fiscal year as a measure of size, return on assets as a measure of profitability, and the logarithm of the banks' age in years. On top of these investor controls, I add bank-specific controls, which are the Tier 1 capital ratio as a measure of riskiness and proxy for being affected by regulatory payout restrictions due to insufficient capital, and the loans to deposits ratio as a proxy for banks' business model. Finally, I also include country-level fixed effects γ_c to account for the existing time-invariant differences in the ownership structure of banks in each country driven by for example its legal framework. Furthermore, I verify that no bank in the sample received bailout money as these often came with dividend restrictions and would thus bias the results if neglected²⁵.

25. I use the information presented by Berger et al. (2025) and Homar (2016) to identify banks which received bailout money

The results are presented in *Table 7*. The estimations in columns (1)-(2) use the SOA measure for dividend smoothing, while columns (3)-(4) use RelVol. Column (1) reveals that the only statistically significant relation is between fund ownership and the dividend yield, as well as the previous years' return volatility. Dividend smoothing is not statistically significantly related to fund ownership, although the coefficient's sign is as expected. Using the additional set of controls in column (2) confirms these findings, although the point estimates are slightly lower and the significance is at the 5% level. Here a one percentage point increase in the dividend yield is related to a 10 % increase in the average aggregate ownership of bank stocks by funds. In the specification with RelVol in columns (3)-(4) there is a statistically significant negative effect at the 10% level for the dividend smoothing proxy RelVol. As in columns (1)-(2), the dividend yield and the previous years' return volatility show positive significant coefficients at least at the 5% level, which are similar in magnitude.

Overall, *Table 7* indicates that fund ownership seems to be driven by the dividend yield rather than by the smoothness of the dividends, because there is no consistent significant effect across the smoothing proxies. This corroborates the previous findings and shows that dividend smoothing seems not to be the main motive for fund investors to invest in banks' stocks. This suggests that the dividend-smoothing channel documented by Larkin, Leary and Michaely (2017) for non-financial firms does not appear to extend to banks.

3.4 Fund-Investor Exposure and Banks' Announcement Returns

If dividend-oriented funds represent an important investor clientele for banks, banks with ex ante greater exposure to such investors may experience stronger stock-price reactions when payout restrictions are announced. I therefore test whether pre-intervention fund ownership predicts announcement-window CARs around the restriction, relaxation, and expiration announcements. This extends the results of previous studies, which primarily examine the average treatment effect of the dividend restrictions (Marsh, 2023; Fringuellotti and Kroen, 2024; Hardy, 2021; Andreeva et al., 2023).

To estimate the effect, I follow the standard event study literature to calculate cumulative abnormal returns for events (see e.g. MacKinlay (1997)) using daily data. The events under study are the announcement date and the two stages where the dividend restrictions were relaxed. To

identify the relevant event dates, I use FactSet’s News 2.0 database to select all news related to the dividend ban²⁶. Furthermore, I include interviews given by SSM officials, which also covered information about relaxations of the dividend restrictions that are available on their webpage²⁷. This strategy reveals that the announcement on the 27th of March 2020 came as a surprise to the public, since I could not identify news about the SSM’s dividend restrictions before the announcement. On the other hand, before the dividend restrictions were officially reduced or completely abandoned, I could identify several news articles and interviews that already indicated a relaxation of the dividend restrictions. The respective dates for the first relaxation in December are as follows: On the 5th of November, Enria pointed towards the wait-and-see strategy regarding the lifting of the restrictions, on the 25th of November 2020, the Financial Times article pointed to the possibility of payouts in 2021, on the 3rd of December 2020 El Confidential mentioned for the first time the limits of the 2021 dividends, and on 15th of December 2020 the official announcement²⁸. For the official abrogation of the policy in 2021 the dates are as follows: On the 15th of June 2021, Enria mentioned that the limitations end soon²⁹, on the 1st of July 2021, Christine Lagarde put in perspective in an ESRB speech the outlook of an end of the measures by September³⁰, and on the 23rd of July 2021 the official announcement.

To calculate abnormal returns, I use daily stock returns from stock prices adjusted for splits and dividends of the 26 banks from the matched sample in Euro. I subtract the one-month Euribor, transformed to daily returns, to get the excess return $R_{b,t}$ over the risk-free rate. The abnormal returns are then defined as:

$$AR_{b,\tau+t} = R_{b,\tau+t} - E[R_{b,\tau+t}] \quad (6)$$

where $AR_{b,\tau+t}$ is the abnormal return of bank b , t days after the event date τ and $E[R_{b,\tau+t}]$ is the expected stock return. For the three policy changes, I set τ to the above-mentioned event dates

26. I filtered the news by the keywords "ECB", "European Central Bank" and "dividend" and used them as relevant sources Street Account, Press Releases, Events, Sector focus News, Crunchbase News, and FactSet Flashwire.

27. See: <https://www.bankingsupervision.europa.eu/press/interviews/html/index.en.html>

28. I focused after the 25th of November only on the news which mentioned a relaxation of the restrictions or included additional information about it.

29. <https://www.bloomberg.com/news/articles/2021-06-15/ecb-hopes-to-lift-bank-dividend-restrictions-soon-enria-says-embedded-checkout=true>

30. <https://www.esrb.europa.eu/news/speeches/date/2021/html/esrb.sp210701-5d14de1059.en.html>

under study³¹.

$E[R_{b,t}]$ is estimated using the Fama-French three factors from Kenneth French's website³².

$$E[R_{b,t}] = \hat{\alpha}_b + \hat{\beta}_b Mkt_t + \hat{\gamma}_b HML_t + \hat{\delta}_b SMB_t \quad (7)$$

where the $\hat{\alpha}_b, \hat{\beta}_b, \hat{\gamma}_b, \hat{\delta}_b$ are the estimated coefficients of the Fama-French three factors model run independently on each stock. Since I am looking at EUR returns and European investors, I transform the US dollar Fama-French factors to EUR factors using the approach of Glück, Hübel and Scholz (2020).

For each event, I use an estimation window of 248 trading days to estimate $E[R_{b,t}]$. For the restriction event the window ends on the 25th of March 2020, for the relaxation event it ends on the 2nd of November 2020, and for the expiration event it ends on the 11th of June 2021. This ensures that the estimation windows are very close to the first announcement date of the three specific events³³. Cumulative abnormal returns $CAR_{b,-1,1}$ for bank b starting one day before and ending one day after the event days mentioned above are then calculated as follows:

$$CAR_{b,-1,1} = \sum_{t=-1}^1 AR_{b,\tau+t} \quad (8)$$

These are then used in a cross-sectional regression where I interact the treatment indicator with an indicator variable that is equal to 1 for high fund ownership to test if there is a differential effect on the valuations:

$$CAR_{b,-1,1} = \alpha + \beta_1 Treated_b + \beta_2 Fund_b + \beta_3 Treated_b \times Fund_b + \beta_4 MktValue_b + \epsilon_b \quad (9)$$

31. To incorporate possible anticipation effects of the news I start the event windows one day before the news announcement. The official introduction and extension of the policy were released after market closing. Therefore, the effective market reaction date (day 0) is the following trading day, meaning the window begins on the calendar announcement date itself (day -1). To ensure full transparency, these exact start dates are printed in the header of *Table 8*

32. For the matched sample of banks during the estimation period, the Capital Asset Pricing Model (CAPM) only achieved an adjusted R^2 of around 0.32 on average using the European market factor of Kenneth French, the three factors model has an adjusted R^2 of around 0.43 on average similar to the five factors model. This indicates that the CAPM might not be a good model to explain stock returns in this period

33. Note that this ensures, for example, that the announcement of the SNB's COVID-19 refinancing facility for banks and the Swiss government's loan guarantee program on the 25th of March are included in the estimation window. These measures would have positively affected Swiss banks' stock returns. See: https://www.snb.ch/en/mmr/reference/pre_20200325/source/pre_20200325.en.pdf

where $Treated_b$ is an indicator variable for being in the treated sample and $Fund_b$ is an indicator variable for the median split share of the percentage of fund owners in February 2020 of the bank, $MktValue_b$ is the market value of the bank in Euro at the beginning of the event.

The results for the different event days are shown in *Table 8*. Panel A shows the estimates for the average treatment effects of the dividend restrictions on banks' stock returns. The announcement of dividend restrictions, on the 27th of March, shows a statistically insignificant negative point estimate. Although the sign of the estimate is expected, this suggests that the dividend restrictions did not have a statistically distinguishable impact on banks' stock returns. Also, the announcements of the relaxation of the policy, where limited dividend payments were allowed, shows no statistically significant impact. Lastly, for the expiration announcement event days, there is no statistically significant impact on the two days where the abrogation of the policy was indicated. However, on the official day of announcement, there is a statistically positive significant effect of 2.1 percentage points in abnormal returns at the 1 % level.

Panel B focuses on the interaction of the funds' ownership share and the average treatment effects. Consistent with the ownership results, the negative valuation effect of the dividend restriction is around 7.5 percentage points larger for treated banks with high fund ownership compared to treated banks with low fund ownership. This implies an average CAR of roughly -3.0 percentage points for high-fund-ownership banks. This suggests that banks with greater ex ante exposure to mutual fund investors experienced stronger negative valuation effects following the dividend restrictions.

During the relaxation phase, no statistically significant differential effect is observed around the early communication events on 5th and 25th of November. However, once the expected payout limits became clearer in the 3rd of December news report and were officially announced on 15th of December, banks with high fund ownership again experienced significantly lower CARs. The estimated differentials amount to approximately -4.7 percentage points on 3 December, translating to an average CAR of roughly -1.5 percentage points for high-fund-ownership banks. On the official announcement date, the differential amounts to -3.2 percentage points, translating into an average CAR of roughly -0.2 percentage points for high-fund-ownership banks. This suggests that investors had expected a more substantial relaxation of the payout restrictions.

Lastly, a similar pattern emerges around the final removal of the restriction. Neither Enria's

interview on 15 June nor Lagarde’s speech on 1 July generates a significant differential effect. In contrast, the official announcement on 23 July is associated with CARs approximately 3 percentage points lower for banks with high fund ownership. Although these banks experience a positive stock-price reaction on average, their smaller response suggests a slower valuation recovery than for other treated banks.

These results remain similar in magnitude but weaker in significance after including the past dividend yield as can be seen in Appendix *Table A7*. This alleviates the concern that higher fund ownership might just proxy for differences in dividend exposure, as the previous results have established a relation between fund ownership and dividend yields. Thus, fund ownership seems to captures information beyond differences in expected dividend payments. Moreover, Appendix *Table A8* replaces the high-fund-ownership indicator with the lagged continuous ownership share. The results remain qualitatively unchanged, with a negative differential for the main policy announcements. At the initial restriction announcement, for example, a one percentage point increase in fund ownership is associated with a 0.3 percentage point lower CAR. Note that the differential is weaker around the final expiration events, where the point estimates remain negative but are no longer statistically significant. Finally, estimating abnormal returns using the Fama-French five-factor model yields virtually identical results (Appendix *Table A9*).

Appendix Figure A7 provides a graphical illustration of the results reported in Table 8. For each event date, the figure plots the residualized announcement date CARs of treated banks against residualized lagged fund ownership. Both variables are residualized with respect to the market value of the banks. The orange line represents the fitted relationship estimated for the control banks. Consistent with the regression results, treated banks with relatively low fund ownership tend to lie above the control benchmark, whereas treated banks with relatively high fund ownership tend to lie below it, illustrating the more negative stock-price response for banks with greater fund ownership.

The results presented so far suggest that there was lower investor demand following the announcement of the dividend restriction for treated banks with more fund owners, reflected in lower CARs for treated banks with a higher share of fund owners. This finding is consistent with the long-term decrease in ownership shares for the matched sample, as the negative CAR differential also persists after the announcement. In particular, the expiration of the policy should have led dividend-focused investors to return to the stock. This pattern holds across treated banks, as re-

flected in the significant positive average effect of 2.1 percentage points. Nevertheless, banks with a high share of fund owners continue to exhibit underperformance even under these circumstances. This is consistent with the persistent reduction in fund ownership documented above, as policy-makers could again intervene in the payout policy, which may discourage funds from reinvesting in these banks.

4 Conclusion

As a response to the 2020 COVID-19 pandemic, regulators around the world restricted banks' dividends to increase banks' resilience. This policy intervention created a laboratory to investigate whether institutional investors of banks' stocks demand (smooth) dividends, a question on which the existing literature provides mixed evidence. In this paper, I analyze how investors in Eurozone banks reacted to the dividend restrictions announced by the SSM in March 2020.

Relying on the quasi-natural experiment set up by the SSM's intervention, I document a decrease in individual funds' ownership shares in treated banks' stocks, whereas other institutional investors do not change their ownership shares. Importantly, because the empirical design nets out general fund flows and pandemic-driven liquidity shocks, this effect represents an active, targeted divestment decision by fund managers rather than mechanical market-wide selling. In the matched sample, individual fund ownership proportionally declines by around 19.5% by the end of the initial restriction period in November 2020 and persists even after dividend payments resume and after the restrictions are fully lifted. The results of the event study also show an exit of funds over time, which is not so persistent. Although the restrictions were temporary, the persistent decline in fund ownership is consistent with the interpretation that investors revised upward the perceived likelihood of future regulatory interventions in banks' payout policies. Supporting this interpretation, funds paying high dividends to their investors or receiving high dividend income from their portfolio holdings react more strongly to the policy. However, I find no evidence that funds passing through their dividend income are driving the results as those do not react more strongly. Further evidence on the pre-intervention determinants of fund ownership indicates that fund ownership is more strongly related to banks' dividend yield than to dividend-smoothing measures. Therefore, the evidence points more strongly to a dividend yield channel than to a dividend smoothing channel

in the spirit of Larkin, Leary and Michaely (2017).

Banks with greater ex ante exposure to fund investors experience lower announcement-window CARs around key payout-restriction announcements. Banks with a higher share of fund owners had a 7.5 percentage points lower treatment effect on their CAR at the initial announcement of the restriction. Similar negative differentials appear around the announcement of limited dividend payouts and, in the indicator specification, around the abrogation announcement, suggesting that fund-investor exposure remained relevant for banks stock price response to dividend restriction announcements.

These findings have important policy implications. At first, I document that the supervisory interventions, even temporary ones, have implications for banks' investor clientele. If dividend-demanding funds represent a substantial share of banks' ownership base, dividend restrictions can be associated to sizeable declines in equity valuations. While the restrictions were intended to strengthen banks' resilience by preserving capital, the accompanying decline in equity valuations is consistent with a higher cost of equity for affected banks. Policymakers should therefore account for the ownership structure of the banking sector when designing payout restrictions and weigh the benefits of higher retained capital against the potential costs associated with lower market valuations.

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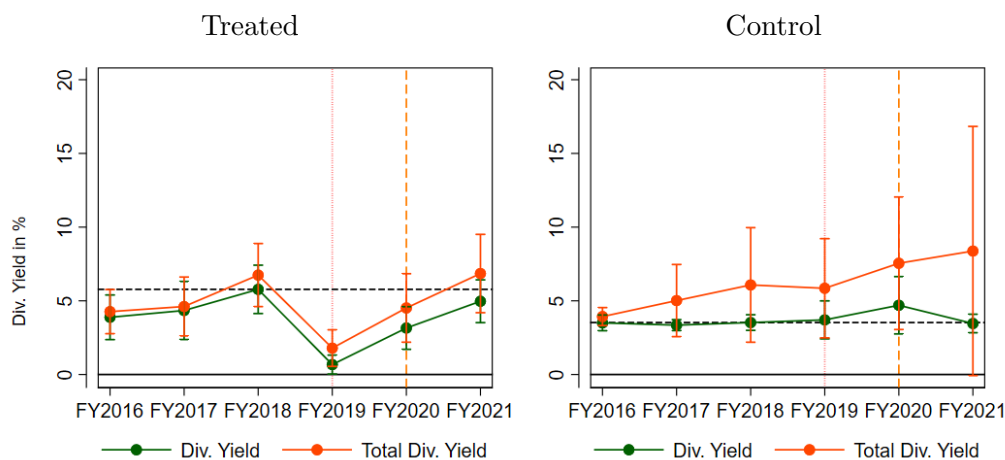
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5 Figures

Figure 1
Dividend Yields of Treated and Control Banks

Figure 1 shows the mean dividend yield in percentage points and dividend payout ratios for the control and treated banks over the time period 2016-2021 for 50 banks. Panel A displays the dividend yield in percentage points, measured by dividends over the fiscal year-end market value of the bank, and the total dividend yield in percentage points, measured by the dividends and repurchases over the fiscal year-end market value of the bank. Panel B displays the dividend payout ratio, measured by DPS over EPS, and the total payout ratio, measured by DPS and Repurchases per outstanding share divided by EPS. Negative payout ratios are excluded. The left-hand side graph displays the values for the treated group, whereas the right-hand side graph displays the values for the control group. The dashed horizontal line indicates the average dividend yield or payout ratio in 2018. The whiskers report the 95% confidence bands of the unconditional means.

Panel A Dividend Yields



Panel B Payout Ratios

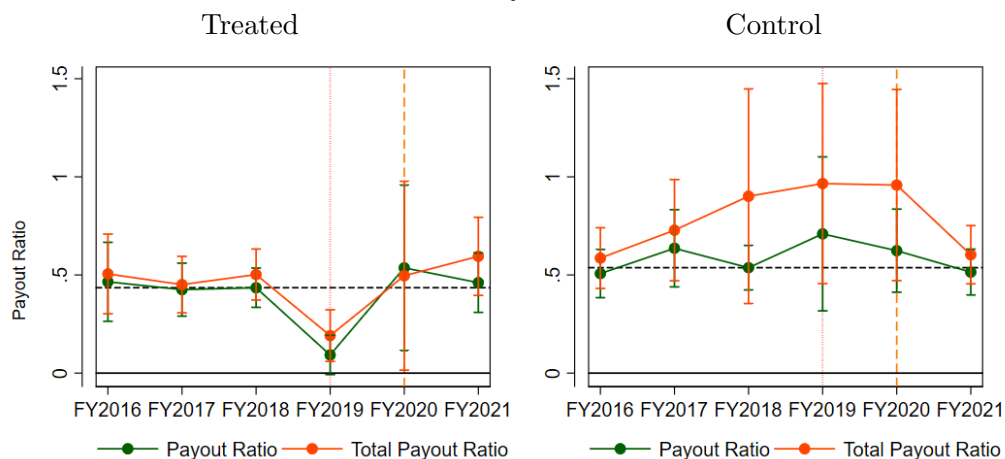


Figure 2**Normalized Bank Dividend Future Response 2020**

Figure 2 Panel A plots the ratio of the EURO STOXX Banks to the EURO STOXX 50 dividend futures across maturities, together with trading volume in the corresponding EURO STOXX Banks dividend futures. Dividend futures are normalized to their respective values on 26 March 2020. Panel B plots an index of the relative cumulative 2020–2022 and 2021–2022 EURO STOXX Banks dividend futures relative to (i) the corresponding cumulative EURO STOXX dividend futures (blue) and (ii) the corresponding cumulative IBES earnings forecasts (grey). The solid red lines indicate the official announcement dates related to the dividend restrictions, the dashed lines indicate ESRB dividend recommendation and the dotted lines indicate news regarding the dividend restriction relaxation.

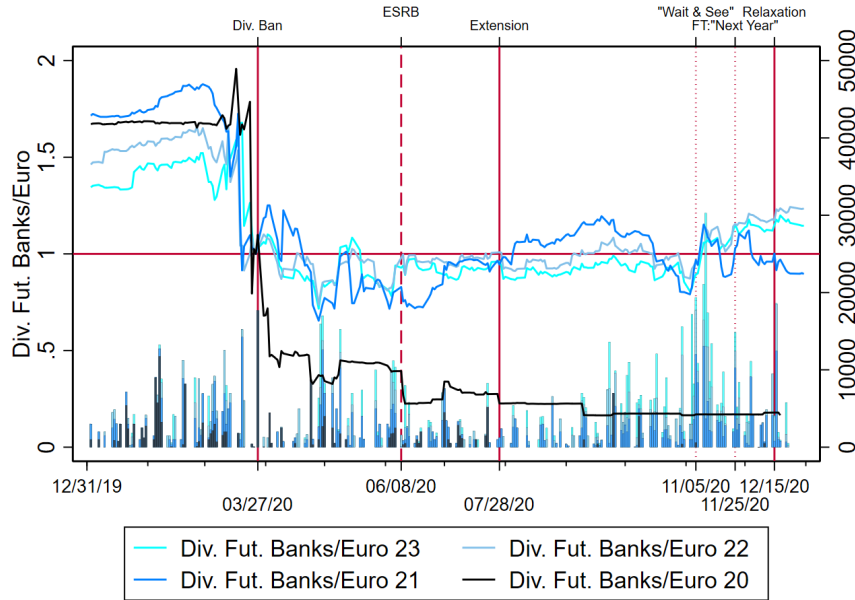
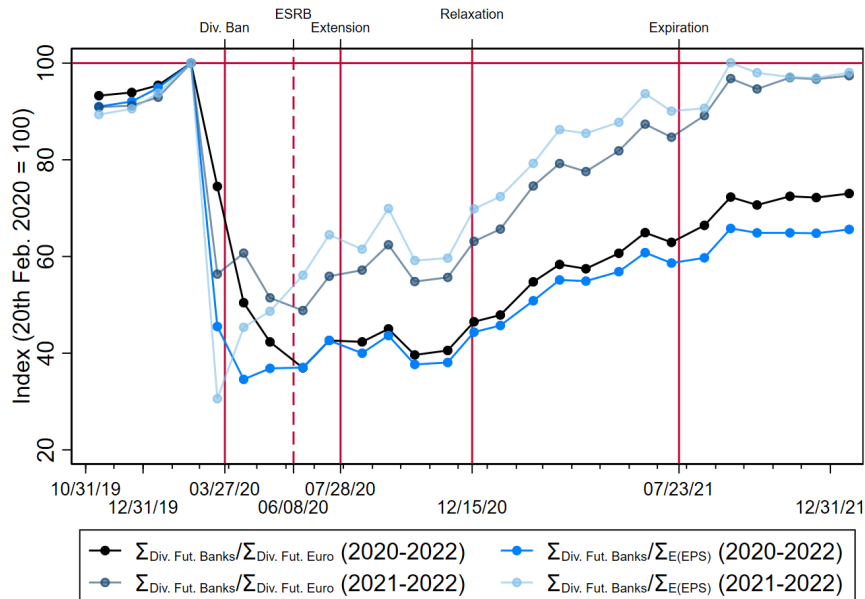
Panel A Dividend Futures**Panel B 2020–2022 Dividend Futures Relative Valuations**

Figure 3**Event Study Plots: Funds and Institutions Matched**

Panels A and B of Figure 3 plot the evolution of the coefficients $\{\hat{\delta}_k\}$ of Equation (2). Panel A and B show the estimates using the poisson pseudo-maximum likelihood regression on $\%Ownership_{i,j,t}$ for funds and institutions on the matched sample. Panel C and D show the estimates using a linear probability model on an indicator variable being one if investor i is invested in bank b at time t for the matched sample. The sample includes 24 banks for funds and 26 banks for institutions. The regression uses bank and $investor \times month$ fixed effects and controls for the exchange rate to EUR, FX to EUR, and the monthly change in shares outstanding per bank $\Delta_t O.S._b$. The outer bands represent the sup-t 95% confidence bands according to Montiel Olea and Plagborg-Møller (2019), the inner bands are the 95% confidence intervals for clustered standard errors on banks. All coefficients measure the impact compared to February 2020. The average value of the dependent variable for the treated group in February 2020 is reported above the coefficient of February 2020. The dotted red line marks the month of the implementation, the dashed orange line marks the relaxation announcement, and the dashed teal line marks the expiration announcement.

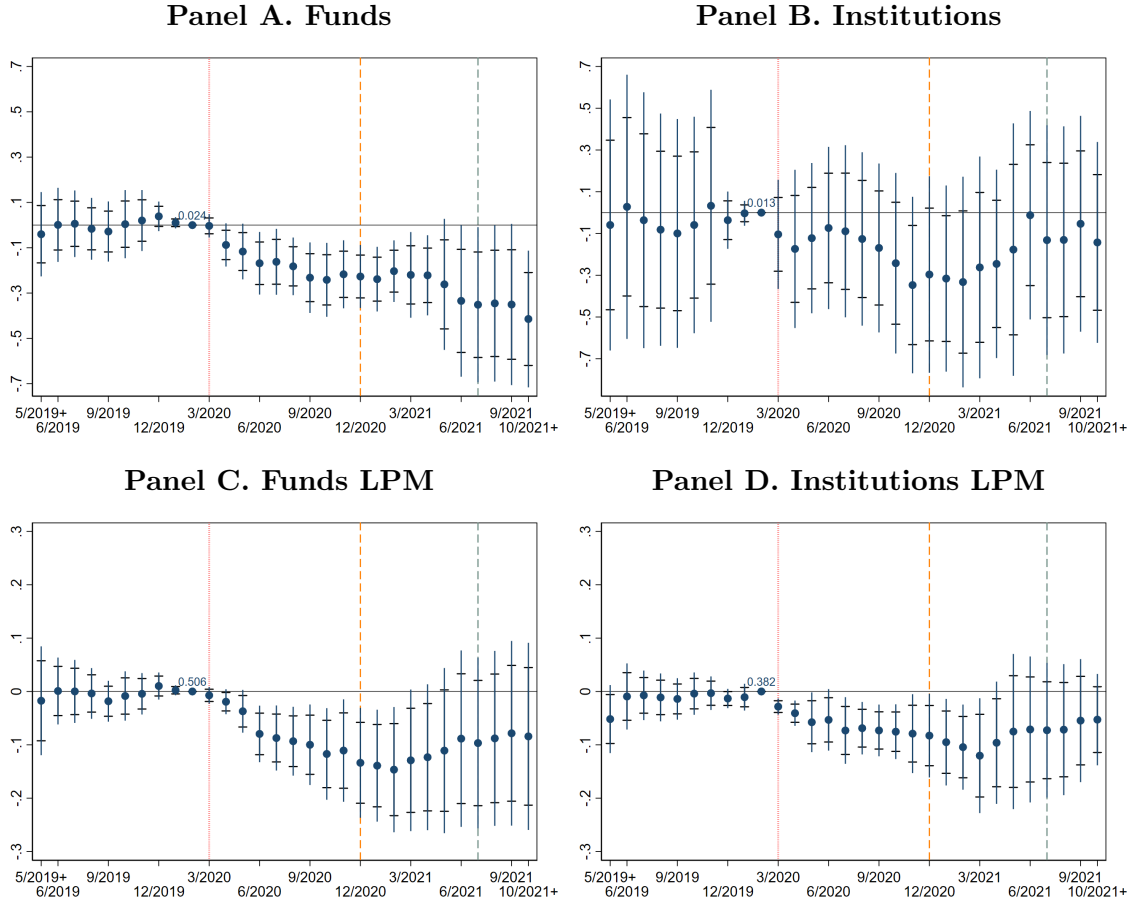


Figure 4
Event Study Plots Investor Dimension

Figure 4 plots the evolution of level estimates of the 4 categories created by the interaction of the two dummy variables, *HighPay* and *HighInc*, in equation 3. The regression uses *investor* $\times$ *month* fixed effects and the bank level controls. The grey series refers to the funds that have low dividend income and low dividend payments, the light blue series refers to funds with only high dividend income, the green series refers to funds that have only high dividend payments and the dark blue series refers to funds with high dividend income and high dividend payments. The outer bands represent the sup-t 95% confidence bands according to Montiel Olea and Plagborg-Møller (2019), the inner bands are the 95% confidence intervals for clustered standard errors on banks. All coefficients measure the impact compared to February 2020. The dotted red line marks the month of the implementation, the dashed orange line marks the relaxation announcement, and the dashed teal line marks the expiration announcement.

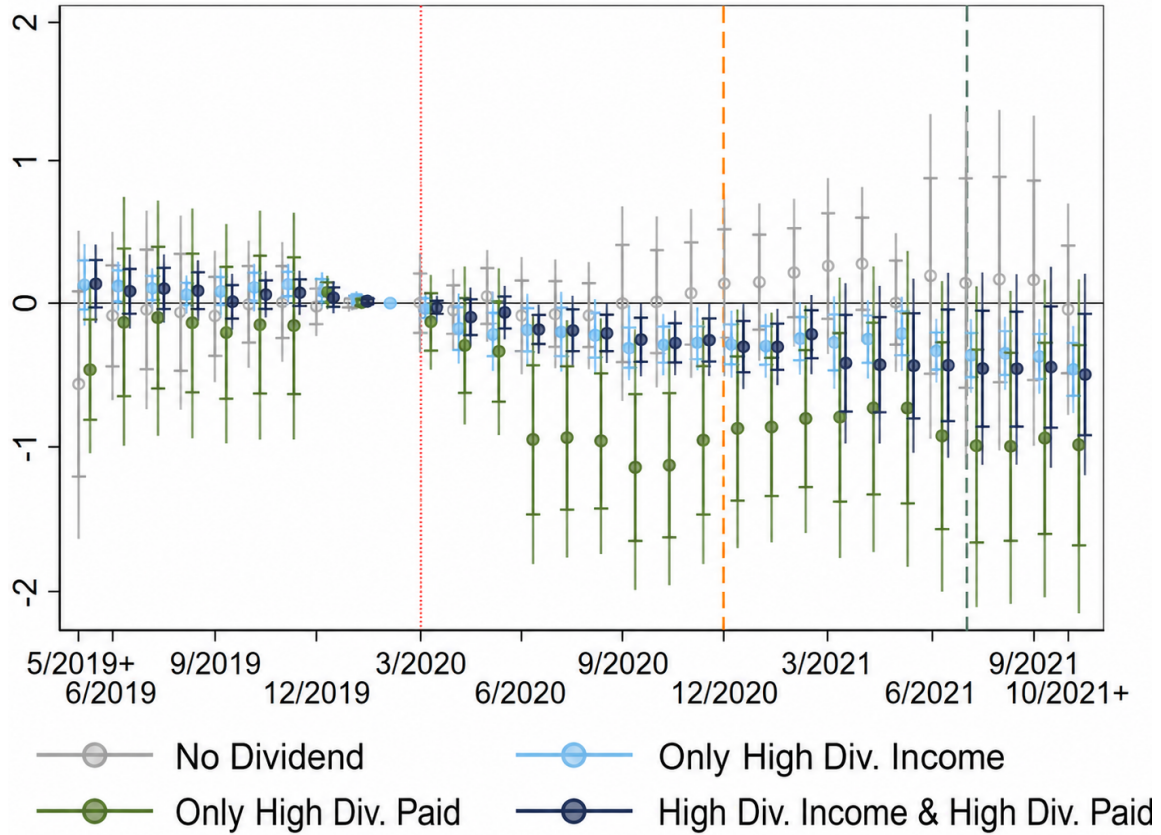
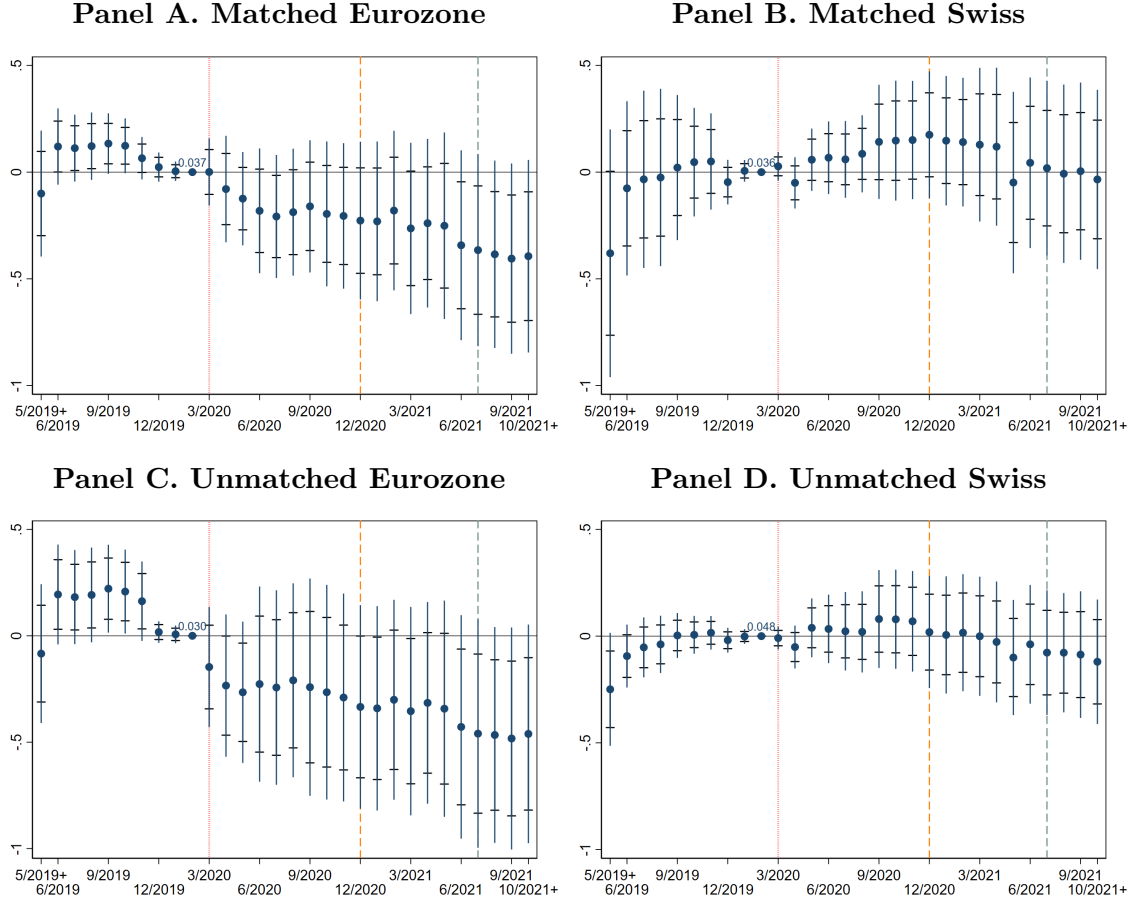


Figure 5**Event Study Plots: EU - Swiss Banks Split**

Panels A, B, C and D of Figure 5 plot the evolution of the coefficients $\{\hat{\delta}_k\}$ of Equation (4) for different samples. Panel A shows the estimates using the matched sample of Eurozone banks, Panel B shows the estimates using the matched sample of Swiss banks, Panel C shows the estimates using the unmatched sample of Eurozone banks, and Panel D shows the estimates using the unmatched sample of Swiss banks. All estimations apply the poisson pseudo-maximum likelihood regression on $\%Ownership_{i,j,t}$ for dividend funds and non-dividend funds. The regression uses investor, $bank \times month$ and $strategy \times month$ fixed effects and controls for the previous month's fund return, the previous month's daily return volatility of the fund, and the previous month's net fund flows (in percentage points). The outer bands represent the sup-t 95% confidence bands according to Montiel Olea and Plagborg-Møller (2019), the inner bands are the 95% confidence intervals for clustered standard errors on banks. All coefficients measure the impact compared to February 2020. The average value of the dependent variable for the treated group in February 2020 is reported above the coefficient of February 2020. The dotted red line marks the month of the implementation, the dashed orange line marks the relaxation announcement, and the dashed teal line marks the expiration announcement.



6 Tables

Table 1
Descriptive Statistics: Annual Data

Table 1 Panel A shows summary statistics for the annual bank-level variables over the time period 2016-2019. Panel B shows the averages of the matched treated and control group in 2019 after applying the propensity score matching. *%Inst.*, *%Fund*, and *%Insider* denote the ownership shares in percentage points of institutional investors, funds, and insiders/stakeholders of the bank, respectively. *Price/Book* is the ratio of the stock price over book equity in percentage points. *Return* is the stock market return over the last year in percentage points. *Div.Yield* is the ratio of dividends per share over the share price in percentage points. Vol.Return_b^d is the daily stock price volatility over 270 business days. *Size(Assets)* is the logarithm of total assets. ROA is net income over average total assets in percentage points. *Log(Age)* is the logarithm of the banks' age. *Tier1 Cap. Ratio* is defined as the total Tier1 capital divided by total risk-weighted assets, both defined by the latest regulatory and supervisory guidelines. *Loans/Deposits* is defined as the ratio of total gross loans over deposits in percentage points. SOA is the speed of adjustment estimate, i.e., β in (A3), winsorized at the 2.5% level. *RelVol* is the relative volatility of DPS to EPS as in (A4), winsorized at the 2.5% level. $WACD_{2019}$ and $WACD_{2020}$ are the yield to maturity of the weighted average cost of debt provided by FactSet in the year 2019 and 2020, respectively. $E(\text{Div.Yield})_{\text{Avg}}$ is the expected dividend yield for 2020 based on analysts dividend forecasts in December 2019. $E(\text{PayoutRatio})_{\text{Avg}}$ is the average expected payout ratio in percentage points for 2020 based on analysts forecasts in December 2019.

Panel A		Summary Statistics				
	mean	sd	min	median	max	count
<i>%Inst.</i>	20.564	16.782	0.000	18.005	65.251	268
<i>%Funds</i>	17.231	14.484	0.000	14.691	47.952	268
<i>%Insider</i>	32.492	28.072	0.000	32.118	99.805	272
<i>Price/Book</i>	94.418	61.985	5.056	78.207	354.740	260
<i>Return</i>	0.120	31.390	-87.760	2.011	198.209	255
<i>Div. Yield</i>	3.146	3.060	0.000	3.019	24.187	258
Vol. Return_b^d	1.874	1.149	0.000	1.591	7.540	261
<i>Size(Assets)</i>	24.703	1.986	18.458	24.773	28.403	360
ROA	0.500	0.734	-3.073	0.427	4.647	359
<i>Log(Age)</i>	4.574	0.828	1.609	4.894	6.304	328
<i>Tier1 Cap. Ratio</i>	18.259	9.936	7.014	16.166	90.855	346
<i>Loans/Deposits</i>	145.608	216.592	27.189	109.572	2428.145	337
SOA	0.542	0.444	-0.173	0.559	1.507	180
<i>RelVol</i>	1.108	1.184	0.078	0.854	6.399	172
Panel B		Post-Trimming Means				
	Diff	p	$Mean_c$	N_c	$Mean_t$	N_t
<i>Price/Book</i>	2.007	0.879	70.841	9	68.833	17
<i>Size(Assets)</i>	-0.722	0.259	24.589	9	25.311	17
<i>Tier1 Cap. Ratio</i>	1.278	0.169	18.006	9	16.728	17
SOA	0.340	0.168	0.786	7	0.446	13
<i>RelVol</i>	0.389	0.230	1.251	7	0.861	12
<i>%Inst.</i>	-11.895	0.121	18.572	9	30.467	17
<i>Loans/Deposits</i>	-2.252	0.909	116.645	9	118.897	17
ROA	-0.094	0.302	0.448	9	0.542	17
$WACD_{2019}$	-0.401	0.559	1.264	9	1.665	15
$WACD_{2020}$	-0.364	0.555	0.847	9	1.211	14
$E(\text{Div.Yield})_{\text{Avg}}$	-1.287	0.082	3.747	7	5.034	15
$E(\text{PayoutRatio})_{\text{Avg}}$	3.785	0.686	52.198	7	48.413	15

Table 2
Descriptive Statistics: Monthly Data

Table 2 shows the mean, standard deviation, 1st quartile, median, 3rd quartile, and the number of observations of the monthly variables used in this study per investor group over the period 2018-2021. % $Own_{b,i}$ is the percentage ownership of each investor. $\mathbb{1}(\%Own_{b,i} > 0)$ is an indicator variable equal to one if investor i is invested in bank b . $Own_{b,i}$ in mn. EUR is the EUR amount invested per bank of each investor in millions. $Return_b^m$ is the monthly stock return of the bank. $Vol.Return_b^d$ is the banks' daily stock return volatility per month. Δ_t O.S. is the change in outstanding shares per bank over time. $FundFlows$ in % are the monthly net flows to net asset value per fund in each month winsorized at the 1 % level. $Fund Size$ in bn. EUR is the total net asset value per fund in billion Euro. $L.Fund Return_i$ is the portfolio return per fund of the previous month. $L.Fund Vol. Return_i^d$ is the daily portfolio return volatility per fund of the previous month. $FundIncome Return$ 2019 in % is the return obtained by the fund's distributions in 2019 that are then reinvested into the fund until the end of the year 2019. $Fund Div.Income$ 2019 in % is the dividends received by fund holdings over 2019 divided by the net asset value of the end of 2018 in percentage points. $Income Return Percentile$ 2019 is the percentile rank relative to all funds that have the same Morningstar fund Category of the funds holding period return that is attributed to dividend distributions in 2019.

		mean	sd	p25	p50	p75	count
Fund	% $Own_{b,i}$	0.022	0.176	0.000	0.001	0.008	1804402
	$\mathbb{1}(\%Own_{b,i} > 0)$	0.574	0.494	0.000	1.000	1.000	1804402
	$Own_{b,i}$ in mn. EUR	3.265	23.358	0.000	0.110	1.313	1790824
	$Return_b^m$	-0.033	11.307	-5.196	0.788	6.231	1802899
	$Vol.Return_b^d$	1.953	1.191	1.252	1.646	2.231	1803190
	$\Delta_t O.S._b$	0.001	0.014	0.000	0.000	0.000	1789405
	$L.FundFlows_i$ in %	0.444	6.153	-1.201	-0.030	1.208	1055869
	$L.Fund Size_i$ in bn. EUR	2.635	15.341	0.161	0.456	1.274	1067919
	$L.Fund Return_i^m$	0.620	4.758	-1.572	0.976	3.190	1099099
	$L.Fund Vol. Return_i^d$	0.737	0.639	0.463	0.620	0.838	1099099
	$Fund Income Return$ 2019 in %	2.009	1.692	0.000	2.085	3.230	868872
	$Fund Div.Income$ 2019 in %	3.569	3.264	2.571	3.322	4.055	911653
	$Income Return Percentile$ 2019	39.576	25.150	22.823	33.000	54.184	871198
Institution	% $Own_{b,i}$	0.020	0.230	0.000	0.000	0.001	721157
	$\mathbb{1}(\%Own_{b,i} > 0)$	0.393	0.488	0.000	0.000	1.000	721157
	$Own_{b,i}$ in mn. EUR	2.669	37.737	0.000	0.000	0.180	715464
	$Return_b^m$	-0.062	11.162	-5.196	0.738	6.154	720148
	$Vol.Return_b^d$	1.935	1.168	1.249	1.640	2.222	720324
	$\Delta_t O.S._b$	0.001	0.015	0.000	0.000	0.000	714921
	$L.FundFlows_i$ in %	-0.720	2.867	-0.960	-0.058	0.091	7178
	$L.Fund Size_i$ in bn. EUR	1.557	2.951	0.129	0.425	1.021	7180
	$L.Fund Return_i^m$	0.604	4.084	-1.223	0.805	2.767	12159
	$L.Fund Vol. Return_i^d$	0.657	0.483	0.399	0.582	0.776	12159
	$Fund Income Return$ 2019 in %	1.172	2.008	0.000	0.823	1.707	4706
	$Fund Div.Income$ 2019 in %	2.356	1.260	1.467	2.107	3.413	5298
	$Income Return Percentile$ 2019	38.924	33.026	12.451	25.000	51.000	4706

Table 3

Matched Sample: Panel Difference in Difference Results Funds

Table 3 shows the results from the difference in difference panel Poisson pseudo-maximum likelihood regressions for different fixed effects on $\%Ownership_{b,i,t}$ for funds. *Restriction* is an indicator equal to one from March 2020 to November 2020. *Relaxation* is an indicator equal to one from December 2020 to June 2021. *Expiration* is an indicator equal to one from July 2021 onwards. *Treated* is an indicator equal to one for banks subject to dividend restrictions. $\Delta_t O.S._b$ is the monthly change in shares outstanding per bank. *FX to EUR* is the exchange rate to EUR for Swiss banks, for other banks, it is 1. $L.Return_b^m$ is the monthly stock return of the previous month of the bank. $L.Vol.Return_b^d$ is the banks' daily stock return volatility of the previous month. Standard errors are clustered on bank. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively. RI p -values for the one-sided alternative hypothesis $H_A : \delta < 0$ are obtained from successful random permutations of the treatment assignment across banks without replacement while keeping the number of treated and control banks constant. Permutations for which the coefficient of interest could not be estimated were excluded.

	(1)	(2)	(3)
<i>Restriction</i> x <i>Treated</i>	0.005 (0.059)	-0.157*** (0.045)	-0.168*** (0.047)
<i>Relaxation</i> x <i>Treated</i>	0.013 (0.080)	-0.232*** (0.069)	-0.236*** (0.067)
<i>Expiration</i> x <i>Treated</i>	0.079 (0.097)	-0.388*** (0.120)	-0.386*** (0.121)
$\Delta_t O.S._b$	-0.536*** (0.124)	-0.829*** (0.158)	-0.816*** (0.157)
<i>FXtoEUR</i>	0.229 (1.192)	-2.999*** (0.854)	-2.911*** (0.854)
$L.Return_b^m$			0.000 (0.001)
$L.Vol.Return_b^d$			0.029 (0.022)
<i>Constant</i>	-3.813*** (1.177)	1.101 (0.840)	0.959 (0.842)
Observations	720,649	586,273	586,273
pseudo R ²	0.050	0.396	0.396
Investor×Month FE		X	X
Bank FE	X	X	X
Month FE	X		
Cluster	Bank	Bank	Bank
# Banks	25	24	24
# FE	47	109,725	109,725
RI p -value, $H_A : \delta_{Restriction \times Treated} < 0$		0.143	0.120
RI p -value, $H_A : \delta_{Relaxation \times Treated} < 0$		0.064	0.058
RI p -value, $H_A : \delta_{Expiration \times Treated} < 0$		0.004	0.006
Random permutations		2000	2000

Table 4

Matched Sample: Panel Difference in Difference Results Institutions

Table 4 shows the results from the difference in difference panel Poisson pseudo-maximum likelihood regressions for different fixed effects on $\%Ownership_{b,i,t}$ for institutions other than funds. *Restriction* is an indicator equal to one from March 2020 to November 2020. *Relaxation* is an indicator equal to one from December 2020 to June 2021. *Expiration* is an indicator equal to one from July 2021 onwards. *Treated* is an indicator equal to one for banks subject to dividend restrictions. $\Delta_t O.S._b$ is the monthly change in shares outstanding per bank. *FX to EUR* is the exchange rate to EUR for Swiss banks, for other banks, it is 1. $L.Return_b^m$ is the monthly stock return of the previous month of the bank. $L.Vol.Return_b^d$ is the banks' daily stock return volatility of the previous month. Standard errors are clustered on bank. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively. RI p -values for the one-sided alternative hypothesis $H_A : \delta < 0$ are obtained from successful random permutations of the treatment assignment across banks without replacement while keeping the number of treated and control banks constant. Permutations for which the coefficient of interest could not be estimated were excluded.

	(1)	(2)	(3)
<i>Restriction</i> $\times$ <i>Treated</i>	0.100 (0.092)	-0.125 (0.163)	-0.173 (0.154)
<i>Relaxation</i> $\times$ <i>Treated</i>	0.208 (0.208)	-0.199 (0.139)	-0.209 (0.138)
<i>Expiration</i> $\times$ <i>Treated</i>	0.326 (0.224)	-0.094 (0.182)	-0.093 (0.182)
$\Delta_t O.S._b$	-0.622* (0.333)	-0.524** (0.258)	-0.494** (0.240)
<i>FXtoEUR</i>	-0.976 (2.350)	-2.404 (2.217)	-2.313 (2.294)
$L.Return_b^m$			-0.000 (0.001)
$L.Vol.Return_b^d$			0.081* (0.043)
<i>Constant</i>	-2.465 (2.260)	1.162 (2.185)	0.921 (2.275)
Observations	249,515	191,074	191,074
pseudo R ²	0.162	0.548	0.548
Investor $\times$ Month FE		X	X
Bank FE	X	X	X
Month FE	X		
Cluster	Bank	Bank	Bank
# Banks	26	26	26
# FE	47	28,471	28,471
RI p -value, $H_A : \beta_{Restriction \times Treated} < 0$		0.325	0.142
RI p -value, $H_A : \beta_{Relaxation \times Treated} < 0$		0.274	0.227
RI p -value, $H_A : \beta_{Expiration \times Treated} < 0$		0.393	0.305
Random permutations		1984	1970

Table 5

Investor Dimension: Panel Triple Difference in Difference Results

Table 5 shows the difference in difference results for the investors' dimension. Columns (1) and (2) show the triple difference panel Poisson pseudo-maximum likelihood regressions on %*Ownership_{b,i,t}* for the matched sample, where column (3) shows the results for the unmatched sample. *D_b* is one for banks subject to dividend restrictions. *Restriction* is an indicator equal to one from March 2020 to November 2020. *Relaxation* is an indicator equal to one from December 2020 to June 2021. *Expiration* is an indicator equal to one from July 2021 onwards. *HighPay* is an indicator for funds with high income return, translating to a below-median Morningstar income-return percentile rank. *HighInc* is an indicator for funds having an above-median dividend income received from their holdings. Standard errors are clustered on bank. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively.

<i>Sample :</i>	Matched	Matched	Unmatched
	(1)	(2)	(3)
<i>Restriction</i> x <i>D_b</i>	0.366 (0.322)		
<i>Relaxation</i> x <i>D_b</i>	0.592* (0.325)		
<i>Expiration</i> x <i>D_b</i>	0.433 (0.526)		
<i>Restriction</i> x <i>D_b</i> x <i>HighPay</i>	-0.803*** (0.222)	-0.952*** (0.252)	-0.591** (0.245)
<i>Relaxation</i> x <i>D_b</i> x <i>HighPay</i>	-1.056*** (0.274)	-1.117*** (0.279)	-0.575** (0.274)
<i>Expiration</i> x <i>D_b</i> x <i>HighPay</i>	-1.092*** (0.380)	-1.038*** (0.368)	0.000 (0.274)
<i>Restriction</i> x <i>D_b</i> x <i>HighInc</i>	-0.732** (0.336)	-0.750** (0.360)	-0.688*** (0.198)
<i>Relaxation</i> x <i>D_b</i> x <i>HighInc</i>	-0.997** (0.398)	-1.018** (0.444)	-0.824*** (0.263)
<i>Expiration</i> x <i>D_b</i> x <i>HighInc</i>	-1.005* (0.564)	-1.007* (0.595)	-0.704*** (0.255)
<i>Restriction</i> x <i>D_b</i> x <i>HighPay</i> x <i>HighInc</i>	0.849*** (0.230)	0.884*** (0.261)	0.719*** (0.238)
<i>Relaxation</i> x <i>D_b</i> x <i>HighPay</i> x <i>HighInc</i>	0.971*** (0.297)	1.047*** (0.315)	0.625** (0.311)
<i>Expiration</i> x <i>D_b</i> x <i>HighPay</i> x <i>HighInc</i>	1.031** (0.475)	1.105** (0.483)	0.174 (0.301)
<i>D_b</i> x <i>HighPay</i>	0.120 (0.170)	0.166 (0.180)	0.105 (0.159)
<i>D_b</i> x <i>HighInc</i>	-0.056 (0.200)	-0.039 (0.202)	0.485* (0.252)
<i>D_b</i> x <i>HighPay</i> x <i>HighInc</i>	-0.165 (0.172)	-0.218 (0.184)	-0.366* (0.218)
Constant	2.547** (1.183)	-1.235*** (0.077)	-1.598*** (0.117)
Observations	226,933	226,933	390,514
pseudo R ²	0.422	0.426	0.413
Bank Controls	Yes		
Bank×Month FE		X	X
Bank FE	X		
Inv.×Month FE	X	X	X

Table 6

Bank Dimension: Panel Triple Difference Results

Table 6 shows the results from the triple difference panel Poisson pseudo-maximum likelihood regressions for different sample splits on $\%Ownership_{b,i,t}$ of funds. Column (1) uses the median split of the smoothing measure SOA, column (2) uses the median split of the smoothing measure RelVol, column (3) uses the median split of the dividend yield, column (4) uses the median split of the Price/Book ratio, column (5) uses the median split of the Log(Z-Score), and column (6) uses the median split of the Tier 1 Cap. Ratio. Note that low SOA and low RelVol indicate more smoothing. *Restriction* is an indicator equal to one from March 2020 to November 2020. *Relaxation* is an indicator equal to one from December 2020 to June 2021. *Expiration* is an indicator equal to one from July 2021 onwards. *Treated* is an indicator equal to one for banks subject to dividend restrictions. Low SOA is an indicator variable for below-median SOA estimates in 2018, i.e., high smoothing. SOA is the speed of adjustment estimate, i.e., β in (A3), winsorized at the 2.5% level. Low RelVol is an indicator variable for below-median RelVol estimates in 2018, i.e., high smoothing. RelVol is the relative volatility of DPS to EPS as in (A4), winsorized at the 2.5% level. *Low Price/Book* is an indicator variable for the below-median Price/Book ratio. *Low Tier1* is an indicator variable for the below-median Tier 1 capital ratio in 2019, i.e., low capitalization. *Low Log(Z - Score)* is an indicator variable for the below-median distance to default in 2019, i.e., high risk. Control variables are the following: $\Delta_t O.S._b$ is the monthly change in shares outstanding per bank. *FX to EUR* is the exchange rate to EUR for Swiss banks, for other banks, it is 1. $L.Return_b^m$ is the monthly stock return of the previous month of the bank. $L.Vol.Return_b^d$ is the banks' daily stock return volatility of the previous month. Standard errors are clustered on bank. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively.

<i>Char:</i>	Low SOA (1)	Low RelVol (2)	Low Div. Yield (3)	Low Price/Book (4)	Low Log(Z-Score) (5)	Low Tier1 (6)
<i>Restriction</i> x <i>Treated</i> x <i>Char</i>	-0.038 (0.096)	-0.177 (0.107)	-0.125 (0.134)	-0.084 (0.161)	0.218** (0.111)	0.106 (0.139)
<i>Relaxation</i> x <i>Treated</i> x <i>Char</i>	0.048 (0.118)	-0.110 (0.119)	-0.205 (0.145)	-0.103 (0.152)	0.053 (0.139)	0.165 (0.160)
<i>Expiration</i> x <i>Treated</i> x <i>Char</i>	0.159 (0.129)	0.091 (0.134)	-0.018 (0.165)	0.020 (0.168)	-0.165 (0.191)	0.249 (0.188)
<i>Restriction</i> x <i>Treated</i>	-0.104** (0.052)	-0.081 (0.052)	-0.099 (0.084)	-0.093 (0.074)	-0.322*** (0.090)	-0.204*** (0.067)
<i>Relaxation</i> x <i>Treated</i>	-0.174*** (0.064)	-0.108** (0.046)	-0.134 (0.094)	-0.134 (0.085)	-0.258** (0.121)	-0.254*** (0.098)
<i>Expiration</i> x <i>Treated</i>	-0.428*** (0.107)	-0.389*** (0.105)	-0.409*** (0.131)	-0.310** (0.130)	-0.253** (0.122)	-0.405*** (0.144)
Observations	421,580	448,268	584,157	586,273	586,273	586,273
Full Controls	Yes	Yes	Yes	Yes	Yes	Yes
Investor×Month FE	X	X	X	X	X	X
Bank FE	X	X	X	X	X	X
Cluster	Bank	Bank	Bank	Bank	Bank	Bank
# Banks	17	18	22	24	24	24

Table 7
Fund Ownership and Dividend Smoothing

Table 7 displays the estimation results of dividend smoothing on fund ownership from 2016 to 2019. % of Funds is the aggregate ownership share of the fund investor group for bank b . SOA is the speed of adjustment estimate, i.e., $\hat{\beta}$ in (A3), winsorized at the 2.5% level. RelVol is the relative volatility of DPS to EPS as in (A4), winsorized at the 2.5% level. Price/Book is the ratio of the stock price over book equity. Return is the stock market return over the last year in percentage points. Div. Yield is the ratio of dividends per share over the market value in percentage points. Size (Assets) is the logarithm of total assets. ROA is defined as net income over average total assets in percentage points. Log(Age) is the logarithm of the banks' age since its inception year. $Vol.Return_b^d$ is the daily stock price volatility over 270 business days. Loans/Deposits is defined as the ratio of total gross loans over deposits. Tier1 Cap. Ratio is defined as the Tier1 capital divided by total risk-weighted assets as defined by the latest regulatory and supervisory guidelines. L. indicates the lag operator. Standard errors are clustered on the bank level. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively.

	(1)	(2)	(3)	(4)
<i>L.SOA</i>	-0.057 (0.171)	-0.048 (0.160)		
<i>L.RelVol</i>			-0.129* (0.074)	-0.130* (0.076)
<i>L.Price/Book</i>	0.000 (0.002)	0.000 (0.002)	-0.000 (0.002)	-0.000 (0.002)
<i>L.Return</i>	0.003 (0.002)	0.002 (0.002)	0.002 (0.002)	0.001 (0.002)
<i>L.Div. Yield</i>	0.109*** (0.041)	0.093** (0.040)	0.107** (0.044)	0.092** (0.043)
<i>L.Vol. Return_b^d</i>	0.179*** (0.056)	0.146** (0.068)	0.175*** (0.064)	0.164** (0.081)
<i>L.Size (Assets)</i>	0.164 (0.102)	0.164 (0.110)	0.168* (0.096)	0.161 (0.100)
<i>L.ROA</i>	0.130 (0.103)	0.127 (0.170)	0.181* (0.100)	0.255 (0.199)
<i>Log(Age)</i>	0.249 (0.237)	0.402 (0.275)	0.225 (0.212)	0.339 (0.262)
<i>L.Tier1 Cap. Ratio</i>		0.010 (0.012)		0.008 (0.014)
<i>L.Loans/Deposits</i>		0.002 (0.002)		0.001 (0.002)
<i>Constant</i>	-3.209 (2.522)	-4.190 (3.094)	-3.085 (2.426)	-3.679 (2.854)
Observations	174	162	165	153
Pseudo R ²	0.526	0.553	0.520	0.541
Cluster	Bank	Bank	Bank	Bank
Country FE	X	X	X	X

Table 8
CAR Regressions: Event Days

Table 8 presents the regressions of equation 9 for the three events under study using the matched sample of 26 banks. Each column shows the CAR for a different event date where the event window is set to $[-1, 1]$ and the start date is displayed in the first row. Abnormal returns are calculated using the Fama-French 3 factor model. The regression controls for the market value, *MktValue*. Standard errors are robust. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively.

	Restriction	Relaxation				Expiration		
Event Window Start:	03/27/20	11/04/20	11/24/20	12/02/20	12/15/20	06/14/21	06/30/21	07/22/21
Panel A: Treatment Effects								
<i>Treated</i>	-1.459 (1.598)	-0.545 (0.938)	-1.418 (1.133)	0.164 (1.032)	1.106 (0.748)	-0.527 (0.532)	1.045 (0.656)	2.113*** (0.726)
<i>Constant</i>	-0.017 (0.951)	-0.228 (0.799)	-0.133 (0.795)	0.344 (0.444)	-0.903** (0.396)	0.366 (0.331)	-0.532 (0.527)	-0.302 (0.516)
N	26	26	26	26	26	26	26	26
adj. R^2	-0.009	-0.064	-0.036	-0.007	0.068	-0.051	0.008	0.155
Controls	X	X	X	X	X	X	X	X
Panel B: Fund Share								
<i>Treated</i>	1.988 (2.273)	-1.217 (1.096)	-0.168 (1.994)	2.924 (1.993)	2.447* (1.183)	-0.443 (0.883)	1.493 (1.268)	3.211*** (0.980)
<i>High Fund Share</i>	3.690 (2.689)	-0.346 (1.989)	-3.076* (1.551)	0.472 (0.957)	2.101** (1.009)	0.049 (0.730)	1.284 (0.764)	2.552** (0.911)
<i>Treated</i> $\times$ <i>High Fund Share</i>	-7.466** (3.403)	1.270 (2.217)	-0.546 (2.282)	-4.773** (2.208)	-3.236** (1.474)	-0.163 (1.095)	-1.319 (1.317)	-2.987** (1.238)
<i>Constant</i>	-1.222 (0.747)	-0.067 (0.861)	0.198 (0.966)	-0.162 (0.352)	-1.471*** (0.329)	0.344 (0.400)	-0.849 (0.655)	-0.959*** (0.316)
N	26	26	26	26	26	26	26	26
adj. R^2	0.077	-0.132	0.085	0.190	0.089	-0.150	-0.047	0.183
Controls	X	X	X	X	X	X	X	X

A Appendices

Appendix A1 Data

Since some listed institutional investors have associated funds in the FactSet ownership data set, I clean the owner-level data for the fund and institution groups using the investor type as follows: First I change the investor group to “fund” if the type either contains the keyword “fund”, “Mutual Fd” or “Private Eq Fd/Alt Invt”. Next, I aggregate for each holder group (i.e., *fund*, *institution*, *insider/stakeholder*) the holder type to reduce the number of types. For example, I aggregate the types “Pension Fund”, “Pension & Life Product” and “Pension Fund Manager” into *Pension Fund*. Given the particular nature of pension and insurance funds, I exclude them from the fund category in the event study. However, this group does not react to dividend restrictions, as evident in Figure A4, which shows the event study results of pension funds.

The aggregated different investor types for the investor group are listed in Table A1.

Special attention has to be placed on institutional investors that also have funds, as their funds’ ownership is included into the institutional ownership. To separate the ownership of funds from other institutional investors, I retrieve for each reported ownership the associated funds of the owner. Next, I match to each reported non-fund owner the associated funds as reported by FactSet and eliminate 13F sub-filers in case an associated fund is connected with two holders. Finally, I subtract the associated funds’ amounts from the institutional owners’ amounts and replace the institutional owners’ amount with zero if the fund amount is larger. This ensures that the institution category captures only institutional investors excluding fund holdings in the monthly data set.

Table A1
Investor Groups Reclassified:

Table A1 shows the different investor types that are classified into the three investor groups: Funds, Other Institutions, and Insiders/Stakeholders

Funds	Other Institutions	Insiders/Stakeholders
Closed-End Fund	Bank Inv. Division	Company
Exchange Traded Funds	Broker	Emp. Stk. Owners. Plan/Trust
Hedge Fund	Family Office	Government
Invest Management Corp.	Foundation/Endowment	Individual
Non-Public Fund	Insurance Company	Joint Venture
Open-End Fund	Insurance Fund	Non-Profit Organization
	Investment Adviser	Subsidiary
	Pension Fund	Venture Capital Private Equity
	Private Banking/Wealth	
	Sovereign Wealth Manager	

Table A2
Descriptive Statistics Dataset: Monthly Data

Table A2 shows the mean, standard deviation, 1st quartile, median, 3rd quartile, and the number of observations of the additional variables describing the investor ownership dataset. US Filings is an indicator being one if the source of the data comes from US filings. Non 13F Based is an indicator if the source of the investor's ownership does not come from 13F filings. Ratio Treated-Control is the ratio of treated banks an investor is invested in compared to all banks in the sample.

	mean	sd	p25	p50	p75	count
Fund						
<i>USFilings_i</i>	0.256	0.437	0.000	0.000	1.000	1804362
<i>non13FBased_i</i>	0.960	0.195	1.000	1.000	1.000	1804402
exposure_treat	54.000	34.680	22.581	63.636	76.923	1036341
Insider						
<i>USFilings_i</i>	0.025	0.157	0.000	0.000	0.000	7052
<i>non13FBased_i</i>	0.992	0.091	1.000	1.000	1.000	21299
exposure_treat	44.396	49.005	0.000	0.000	100.000	15290
<i>Institutional Share Class Fraction in %</i>						
<i>USFilings_i</i>	0.387	0.487	0.000	0.000	1.000	720976
<i>non13FBased_i</i>	0.614	0.487	0.000	1.000	1.000	721157
exposure_treat	55.800	34.075	40.000	65.217	77.778	283303
Total						
<i>USFilings_i</i>	0.293	0.455	0.000	0.000	1.000	2532390
<i>non13FBased_i</i>	0.862	0.344	1.000	1.000	1.000	2546858
exposure_treat	54.272	34.775	22.581	63.636	77.778	1334934

Table A3
List of Sample Banks

Table A3 lists the banks in the sample for the event study analysis and the matched sample. Banks marked with $\star$ were omitted in the event study due to missing variables and ensuring a balanced panel.

	Full sample	Matched sample	Omitted
AIB Group PLC	Treated	Treated	
Aareal Bank AG	Treated	Treated	
BAWAG Group AG	Treated	Treated	
BNP Paribas S.A. Class A	Treated		
BPER Banca S.p.A.	Treated	Treated	
Banca Popolare di Sondrio S.c.p.A.	Treated	Treated	
Banco BPM SpA	Treated	Treated	
Banco Santander, S.A.	Treated		
Bank of Ireland Group Plc	Treated	Treated	
Bank of Valletta P.L.C.	Treated	Treated	
Commerzbank AG	Treated		
Credit Agricole SA	Treated		
Deutsche Pfandbriefbank AG	Treated	Treated	
HSBC Bank Malta P.L.C.	Treated	Treated	
ING Groep NV	Treated	Treated	
Intesa Sanpaolo S.p.A.	Treated	Treated	
KBC Group N.V.	Treated	Treated	
Liberbank SA	Treated		$\star$
Mediobanca - Banca di Credito Finanziario S.p.A.	Treated	Treated	
Nordea Bank Abp	Treated		$\star$
Nova Ljubljanska banka d.d.	Treated		$\star$
Raiffeisen Bank International AG	Treated	Treated	
Societe Generale S.A. Class A	Treated		
UniCredit S.p.A.	Treated	Treated	
Unicaja Banco S.A.	Treated	Treated	
Vseobecna uverova banka, a.s.	Treated		$\star$
Banque Cantonale Vaudoise	Control		
Banque Cantonale de Bale Campagne Kantonalkbank-Zertifikat	Control		
Banque Cantonale de Geneve SA	Control	Control	
Banque Cantonale du Jura	Control	Control	
Basler Kantonalkbank Partizipsch	Control		
Bellevue Group AG	Control		
Berner Kantonalkbank AG	Control	Control	
Cembra Money Bank AG	Control		
Credit Suisse Group AG	Control	Control	
EFG International AG	Control		
Glarner Kantonalkbank	Control		
Graubuendner Kantonalkbank	Control		
Hypothekarbank Lenzburg AG	Control		
Julius Baer Gruppe AG	Control		
Luzerner Kantonalkbank AG	Control	Control	
ONE swiss bank SA	Control		
St.Galler Kantonalkbank AG	Control	Control	
Swissquote Group Holding Ltd.	Control		
Thurgauer Kantonalkbank	Control	Control	
UBS Group AG	Control	Control	
Valiant Holding AG	Control	Control	
Vontobel Holding AG	Control		
Walliser Kantonalkbank	Control		
Zuger Kantonalkbank AG	Control		

Figure A1
Eurozone Investor Data Coverage

Figure A1 plots the aggregate investment of Eurozone investors, excluding pension funds and insurance corporations, in listed shares of banks for different datasets. The green line relies on the SHS dataset of the ECB, the orange line relies on data retrieved from FactSet and used in the study, and the blue line relies on data retrieved from FactSet excluding 13F sources that is used in Section 3.2.

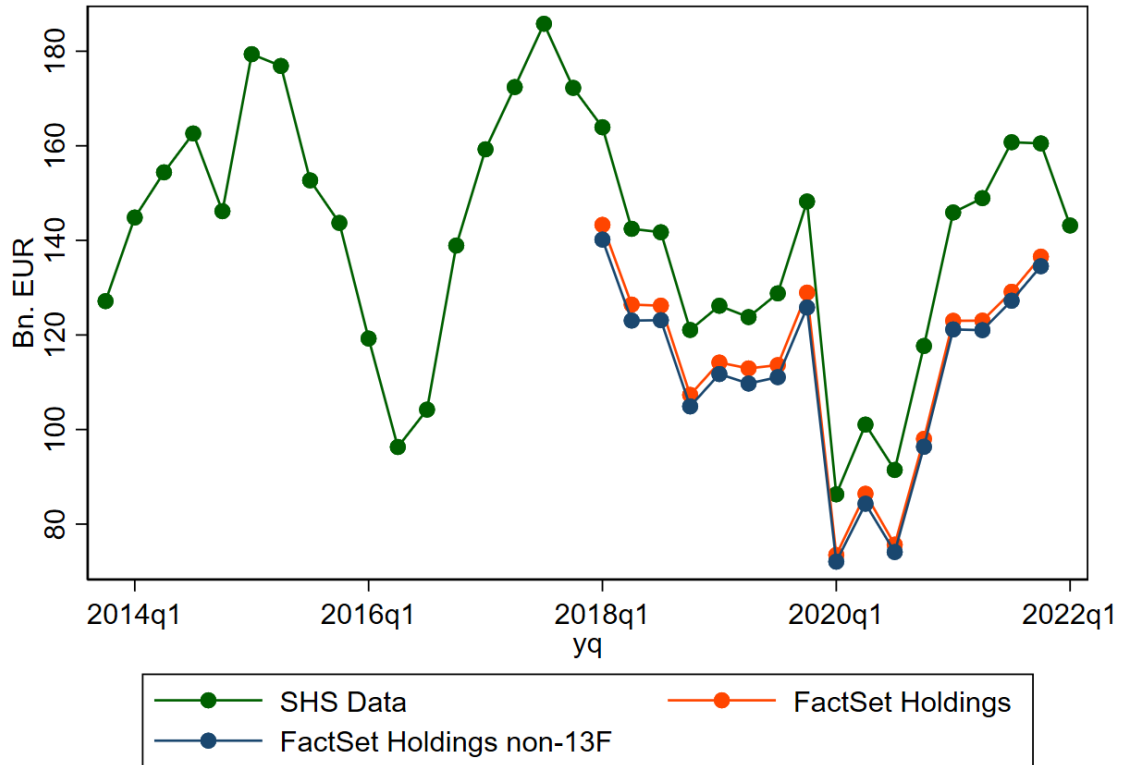
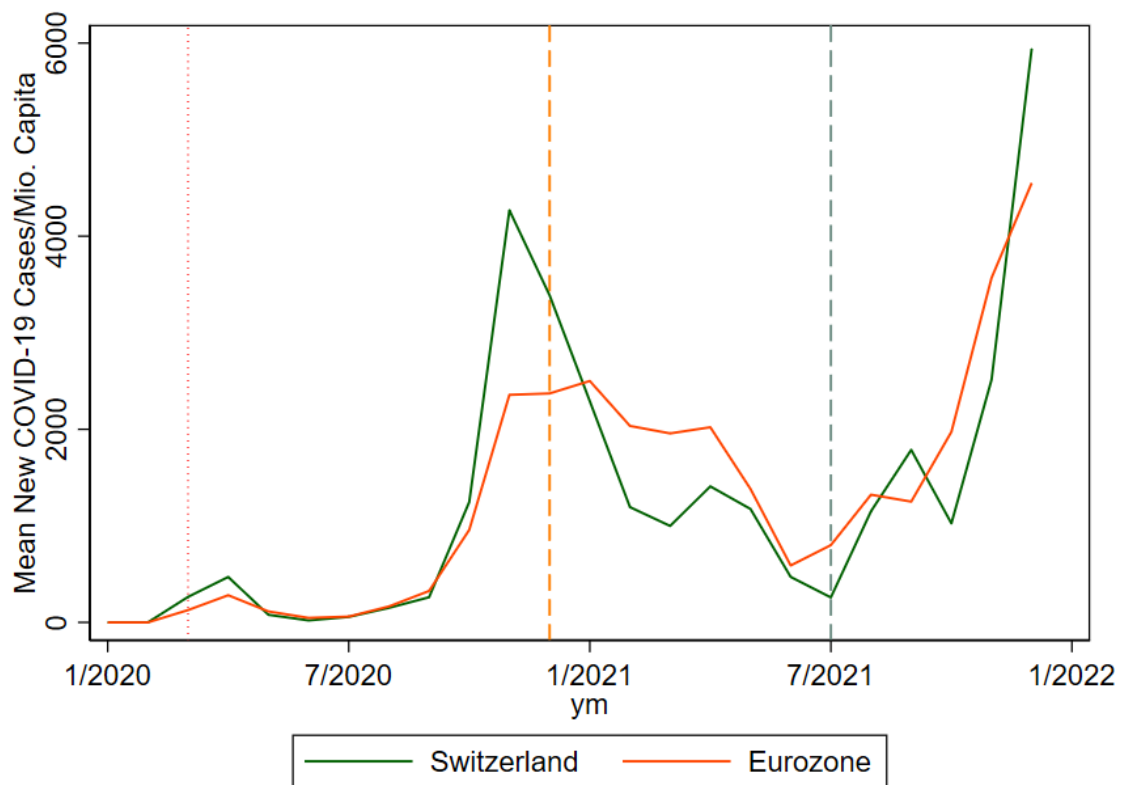


Figure A2
COVID-19 Cases Eurozone and Switzerland

Figure A2 plots the 7-day smoothed new confirmed COVID-19 cases per million capita for the 19 Eurozone countries and Switzerland. Depicted is always the end of month data. The data is taken from the 'Our World in Data' database available under <https://ourworldindata.org/>.



Appendix A2 Morningstar Data

To establish the results for the mechanism, I rely on Morningstar data for a subset of the data set. Unfortunately, these data are only available for a subset of investors, as I could not retrieve an ISIN or CUSIP identifier from FactSet for all securities required to match the datasets. In this section, I will describe how I constructed the dividend income variable and also present some results about potential selection of this subsample.

Appendix A2.1 Dividend Income

For the matched sample of funds that are in Morningstar and the FactSet database, I retrieve the monthly fund holdings over the period 2019 to 2022. I then match the fund holdings using first the ISIN, followed by CUSIP, TICKER and name combinations, and lastly names which are manually verified with Compustat Global and the Compustat North America database. From the latter database, the monthly dividend is sourced. Since some funds do not report on a monthly basis, I interpolate holdings between observed reporting dates. To calculate the dividends received over the year 2019 from the holdings, I assume that between two month end dates, the holdings are linearly adjusted and thus attribute the midpoint to the dividend in data month. The data is then aggregated over the year to capture received dividends in 2019 and then divided by the net asset value at the end of 2018. In particular, it is defined by:

$$DividendIncome_i^H = \frac{\sum_{t=m-11}^m \sum_j s_{jit} div_{jt}}{nav_{i,m-12}} \quad (A1)$$

, where m is the month at the end of the year, i is the fund, j is the stock, s_{jit} is the average of the stocks held between period t and $t - 1$, div_{jt} are the dividends paid per stock in month t , and $nav_{i,m-12}$ is the net asset value of the fund at the end of the previous year. That the interpolation assumption does not change the results, I test if the derived dividend variables are significantly different from each other if I would have just carried forward the last observed variable, and both dividend income as well as the median split categories are not significantly different from each other. This alleviates concerns about the interpolation assumption.

Appendix A2.2 Missing Morningstar Variables

Another potential problem is that the dividend income and dividend return ranking are only available for a subset of the datasets. This could potentially lead to the fact that the mechanism is shown for a selected sample and thus might not be meaningful for the results established in *Table 3*. Below I used the sample of *Table 3* and interacted the treatment dummy with an indicator variable if the Morningstar variable was missing. *Table A4* shows the results of this estimation for the different Morningstar variables. Across the specifications the negative treatment effect is present in the sample where the data is available. However, looking at the differential effect for the sample where the data is missing, one can see that in most specifications the coefficients are insignificant. If anything, the interaction coefficients are positive and thus would indicate that the funds for which this variable is not available are the funds that are less reacting to the restriction.

Table A4
Selection Test:

Table A4 reestimates the specification of *Table 3* with an additional dummy variable equal to one if the Morningstar data is missing interacted with the treatment effect. The dependent variable is $\%Ownershare_{b,i,t}$. *Restriction* is an indicator equal to one from March 2020 to November 2020. *Relaxation* is an indicator equal to one from December 2020 to June 2021. *Expiration* is an indicator equal to one from July 2021 onwards. *Treated* is an indicator equal to one for banks subject to dividend restrictions. *Missing* is an indicator equal to one for funds where the Morningstar variable is missing. $\Delta_t O.S._b$ is the monthly change in shares outstanding per bank. *FX to EUR* is the exchange rate to EUR for Swiss banks, for other banks, it is 1. $L.Return_b^m$ is the monthly stock return of the previous month of the bank. $L.Vol.Return_b^d$ is the banks' daily stock return volatility of the previous month. Standard errors are clustered on bank. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively.

<i>Missing:</i>	Both (1)	Dividend Income (2)	Income Return Rank (3)
Restriction x Treated	-0.185*** (0.057)	-0.160*** (0.051)	-0.182*** (0.054)
Relaxation x Treated	-0.276*** (0.070)	-0.252*** (0.071)	-0.272*** (0.075)
Expiration x Treated	-0.435*** (0.138)	-0.397*** (0.145)	-0.447*** (0.136)
Restriction x Treated x Missing	0.079 (0.112)	-0.022 (0.072)	0.080 (0.101)
Relaxation x Treated x Missing	0.153 (0.096)	0.085 (0.102)	0.160* (0.089)
Expiration x Treated x Missing	0.185 (0.115)	0.074 (0.134)	0.255* (0.136)
Treated x Missing	-0.172** (0.072)	-0.161 (0.106)	-0.188** (0.073)
$\Delta_t O.S._b$	-0.818*** (0.158)	-0.817*** (0.158)	-0.817*** (0.158)
FX to EUR	-2.871*** (0.844)	-2.861*** (0.856)	-2.869*** (0.844)
$L.Return_b^m$	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
$L.Vol.Return_b^d$	0.029 (0.022)	0.029 (0.022)	0.029 (0.022)
Constant	0.963 (0.837)	0.944 (0.838)	0.959 (0.833)
Observations	586,273	586,273	586,273
pseudo R ²	0.396	0.396	0.396
FE	InvestorMonth Bank	InvestorMonth Bank	InvestorMonth Bank
Cluster	Bank	Bank	Bank
# Banks	24	24	24
# FE	109,725	109,725	109,725

Appendix A3 Dividend Smoothing Measures

To measure the dividend smoothing of European banks I rely on the approach of Leary and Michaely (2011). They show that their two measures of dividend smoothing, i.e., speed of adjustment (SOA) and relative volatility (RelVol), can partially offset the small sample bias which is usually an issue for the SOA using the approach of Lintner (1956). Given the short time horizon under study to avoid crisis periods, such an adjustment is necessary for the analysis.

SOA according to Leary and Michaely (2011) is very similar to the classical partial adjustment model of Lintner (1956), where in this case a two-step approach is used. So SOA is defined in a two-step approach according to the following formulae:

$$dev_{i,t} = TPR_{i,t}EPS_{i,t} - D_{i,t-1} \quad (A2)$$

$$\Delta D_{i,t} = \alpha + \beta dev_{i,t} + \epsilon_{i,t}, \quad (A3)$$

where $D_{i,t}$ is dividends per share (DPS) of bank i at time t , $EPS_{i,t}$ is the earnings per share (EPS) of bank i at time t , $TPR_{i,t}$ is the target payout ratio of bank i at time t , and $\epsilon_{i,t}$ is the error term. In the first stage, an estimate of the target payout ratio is needed. This is captured by $TPR_{i,t}$, which is calculated as the median of the payout ratio, i.e., DPS over EPS, from $t - 4$ to t^{34} . In the second stage, equation (A3) is then estimated using rolling regressions, to receive an estimate of the adjustment of the target payout ratio to changes in dividends β . Using dividends per share in the target payout ratio is in line with the finding that dividends per share are an important proxy for payout policy Brav et al. (2005).

The alternative non-parametric measure of dividend smoothing used by Leary and Michaely (2011) is RelVol which is defined as follows:

$$RelVol = \frac{\sigma_{\eta_{i,t}^d}}{\sigma_{\eta_{i,t}^e}} \quad (A4)$$

where $\sigma_{\eta_{i,t}^d}$ and $\sigma_{\eta_{i,t}^e}$ are the root mean squared errors of the respective quadratic time trend estimations on DPS and targeted earnings based dividends:

$$D_{i,t} = \alpha_d + \beta_d t + \gamma_d t^2 + \eta_{i,t}^d, \quad (A5)$$

$$TPR_{i,t}EPS_{i,t} = \alpha_e + \beta_e t + \gamma_e t^2 + \eta_{i,t}^e \quad (A6)$$

Therefore, RelVol measures how volatile dividends are relative to the target dividends. These two measures both capture dividend smoothing, but different parts of it. Whereas RelVol captures the overall volatility of dividends relative to their earnings-implied target, SOA captures how quickly management adjusts the dividend to reach that target payout. Therefore, an investor who prioritizes a highly stable, predictable cash dividend across time, preferring that management smooth right through short-term earnings shocks, would look for a slower adjustment speed and thus a lower SOA.

To obtain the measures SOA and RelVol I estimate the rolling window regressions of

34. Leary and Michaely (2011) highlight in a footnote that using only five years for the estimation of TPR instead of ten does not alter the results

equations (A3), (A5), (A6) using an eight-year window. Similar to Leary and Michaely (2011) I drop observations in the sample where the banks did not yet initiate dividends (i.e., the first observations with zero dividends) and when banks stopped paying dividends (i.e., the last observations with zero dividends). Furthermore, I dropped observations where the TPR was negative and when banks did not pay any dividend in the estimation window. These two measures are then also winsorized at the 2.5% level to eliminate outliers.

Appendix A4 Additional Results

Table A5
Event Study Regressions: Funds

Table A5 shows the results from the panel event study regressions for the Poisson pseudo-maximum likelihood regression and the linear probability model for funds. Only the interaction term coefficients and the controls are reported. $\Delta_t O.S._b$ is the monthly change in shares outstanding per bank. $FX \text{ to } EUR$ is the exchange rate to EUR for Swiss banks, for other banks, it is 1. $L.Return_b^m$ is the monthly stock return of the previous month of the bank. $L.Vol.Return_b^d$ is the banks' daily stock return volatility of the previous month. Standard errors are clustered on banks. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively.

	$\%Own.b,i$	$\mathbb{1}(\%Own_{b,i} > 0)$	$\%Own.b,i$	$\mathbb{1}(\%Own_{b,i} > 0)$
<i>pre 2019m6 $\times$ Treated</i>	-0.0403 (0.0645)	-0.0174 (0.0363)	-0.0718 (0.0738)	-0.0065 (0.0374)
<i>2019m6 $\times$ Treated</i>	0.0010 (0.0567)	0.0009 (0.0223)	-0.0234 (0.0600)	0.0087 (0.0234)
<i>2019m7 $\times$ Treated</i>	0.0059 (0.0509)	0.0001 (0.0210)	-0.0153 (0.0556)	0.0074 (0.0203)
<i>2019m8 $\times$ Treated</i>	-0.0164 (0.0473)	-0.0037 (0.0169)	-0.0377 (0.0500)	0.0028 (0.0168)
<i>2019m9 $\times$ Treated</i>	-0.0284 (0.0460)	-0.0183 (0.0136)	-0.0630 (0.0545)	-0.0079 (0.0147)
<i>2019m10 $\times$ Treated</i>	0.0042 (0.0522)	-0.0085 (0.0164)	-0.0225 (0.0542)	0.0011 (0.0183)
<i>2019m11 $\times$ Treated</i>	0.0201 (0.0467)	-0.0043 (0.0138)	-0.0074 (0.0487)	0.0062 (0.0168)
<i>2019m12 $\times$ Treated</i>	0.0386* (0.0227)	0.0102 (0.0090)	0.0170 (0.0274)	0.0168 (0.0103)
<i>2020m1 $\times$ Treated</i>	0.0102 (0.0086)	0.0022 (0.0033)	0.0016 (0.0113)	0.0061 (0.0050)
<i>2020m3 $\times$ Treated</i>	-0.0034 (0.0178)	-0.0072 (0.0055)	-0.0173 (0.0215)	-0.0033 (0.0063)
<i>2020m4 $\times$ Treated</i>	-0.0876*** (0.0331)	-0.0192** (0.0083)	-0.1273** (0.0550)	-0.0115 (0.0104)
<i>2020m5 $\times$ Treated</i>	-0.1167*** (0.0426)	-0.0371** (0.0143)	-0.1447*** (0.0549)	-0.0284** (0.0136)
<i>2020m6 $\times$ Treated</i>	-0.1685*** (0.0479)	-0.0795*** (0.0188)	-0.1938*** (0.0607)	-0.0706*** (0.0148)
<i>2020m7 $\times$ Treated</i>	-0.1620*** (0.0505)	-0.0872*** (0.0217)	-0.2064*** (0.0666)	-0.0735*** (0.0153)
<i>2020m8 $\times$ Treated</i>	-0.1821*** (0.0441)	-0.0932*** (0.0229)	-0.2155*** (0.0526)	-0.0814*** (0.0189)
<i>2020m9 $\times$ Treated</i>	-0.2321*** (0.0540)	-0.0999*** (0.0268)	-0.2725*** (0.0706)	-0.0855*** (0.0219)

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Table A5 – continued from previous page

	$\%Own.b,i$	$\mathbb{1}(\%Own_{b,i} > 0)$	$\%Own.b,i$	$\mathbb{1}(\%Own_{b,i} > 0)$
$2020m10 \times Treated$	-0.2417*** (0.0566)	-0.1171*** (0.0305)	-0.2487*** (0.0574)	-0.1165*** (0.0295)
$2020m11 \times Treated$	-0.2171*** (0.0520)	-0.1107*** (0.0341)	-0.2498*** (0.0632)	-0.1006*** (0.0305)
$2020m12 \times Treated$	-0.2269*** (0.0482)	-0.1337*** (0.0366)	-0.2836*** (0.0662)	-0.1138*** (0.0341)
$2021m1 \times Treated$	-0.2386*** (0.0495)	-0.1391*** (0.0373)	-0.2685*** (0.0535)	-0.1286*** (0.0341)
$2021m2 \times Treated$	-0.2033*** (0.0471)	-0.1465*** (0.0417)	-0.2308*** (0.0521)	-0.1387*** (0.0386)
$2021m3 \times Treated$	-0.2198*** (0.0656)	-0.1291** (0.0472)	-0.2557*** (0.0620)	-0.1162** (0.0455)
$2021m4 \times Treated$	-0.2216*** (0.0613)	-0.1233** (0.0486)	-0.2277*** (0.0538)	-0.1179** (0.0472)
$2021m5 \times Treated$	-0.2617*** (0.1003)	-0.1108* (0.0550)	-0.2774*** (0.0971)	-0.1046* (0.0531)
$2021m6 \times Treated$	-0.3345*** (0.1161)	-0.0884 (0.0588)	-0.3521*** (0.1120)	-0.0821 (0.0581)
$2021m7 \times Treated$	-0.3514*** (0.1187)	-0.0967 (0.0568)	-0.3672*** (0.1191)	-0.0906 (0.0537)
$2021m8 \times Treated$	-0.3455*** (0.1197)	-0.0879 (0.0583)	-0.3677*** (0.1145)	-0.0810 (0.0585)
$2021m9 \times Treated$	-0.3505*** (0.1233)	-0.0784 (0.0615)	-0.3753*** (0.1206)	-0.0695 (0.0588)
$\Delta_t O.S_b$	-0.8272*** (0.1621)	0.0159 (0.0465)	-0.8105*** (0.1614)	0.0077 (0.0443)
$FX \text{ to } EUR$	-2.4021*** (0.8916)	-0.4756 (0.3808)	-2.1035** (0.9279)	-0.5697 (0.3877)
$L.Return_b^m$			0.0003 (0.0005)	-0.0003 (0.0003)
$L.Vol.Return_b^d$			0.0297 (0.0230)	-0.0094 (0.0070)
$post\ 2021m9 \times Treated$	-0.4145*** (0.1046)	-0.0842 (0.0624)	-0.4215*** (0.1032)	-0.0814 (0.0613)
$Constant$	0.5269 (0.8572)	1.0204*** (0.3586)	0.1963 (0.9040)	1.1240*** (0.3698)
Observations	586,273	672,949	586,273	672,949
adj.R ²		0.304		0.304
Banks	24	24	24	24
$Investor \times Month$ FE	X	X	X	X
$Bank$ FE	X	X	X	X

Table A6
Event Study Regressions: Institutions

Table A6 shows the results from the panel event study regressions for the Poisson pseudo-maximum likelihood regression and the linear probability model for institutions. Only the interaction term coefficients and the controls are reported. $\Delta_t O.S._b$ is the monthly change in shares outstanding per bank. FX to EUR is the exchange rate to EUR for Swiss banks, for other banks, it is 1. $L.Return_b^m$ is the monthly stock return of the previous month of the bank. $L.Vol.Return_b^d$ is the banks' daily stock return volatility of the previous month. Standard errors are clustered on banks. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively.

	$\%Own_{b,i}$	$\mathbb{1}(\%Own_{b,i} > 0)$	$\%Own_{b,i}$	$\mathbb{1}(\%Own_{b,i} > 0)$
<i>pre 2019m6 $\times$ Treated</i>	-0.0591 (0.2072)	-0.0517** (0.0223)	-0.1226 (0.2059)	-0.0492** (0.0235)
<i>2019m6 $\times$ Treated</i>	0.0279 (0.2181)	-0.0094 (0.0217)	-0.0294 (0.2190)	-0.0071 (0.0229)
<i>2019m7 $\times$ Treated</i>	-0.0363 (0.2112)	-0.0073 (0.0162)	-0.0754 (0.2192)	-0.0057 (0.0170)
<i>2019m8 $\times$ Treated</i>	-0.0817 (0.1917)	-0.0111 (0.0157)	-0.1233 (0.1952)	-0.0097 (0.0164)
<i>2019m9 $\times$ Treated</i>	-0.0996 (0.1889)	-0.0140 (0.0136)	-0.1851 (0.2082)	-0.0111 (0.0153)
<i>2019m10 $\times$ Treated</i>	-0.0592 (0.1787)	-0.0040 (0.0139)	-0.1067 (0.1848)	-0.0024 (0.0151)
<i>2019m11 $\times$ Treated</i>	0.0326 (0.1915)	-0.0032 (0.0110)	-0.0103 (0.1902)	-0.0017 (0.0124)
<i>2019m12 $\times$ Treated</i>	-0.0363 (0.0473)	-0.0132** (0.0063)	-0.0748 (0.0556)	-0.0118 (0.0069)
<i>2020m1 $\times$ Treated</i>	-0.0033 (0.0206)	-0.0107 (0.0088)	-0.0083 (0.0205)	-0.0102 (0.0088)
<i>2020m3 $\times$ Treated</i>	-0.1041 (0.0901)	-0.0283*** (0.0053)	-0.1500* (0.0844)	-0.0276*** (0.0057)
<i>2020m4 $\times$ Treated</i>	-0.1738 (0.1306)	-0.0407*** (0.0083)	-0.3480** (0.1478)	-0.0360*** (0.0098)
<i>2020m5 $\times$ Treated</i>	-0.1222 (0.1240)	-0.0577*** (0.0196)	-0.1865 (0.1334)	-0.0552*** (0.0192)
<i>2020m6 $\times$ Treated</i>	-0.0736 (0.1339)	-0.0532** (0.0201)	-0.1360 (0.1447)	-0.0517** (0.0192)
<i>2020m7 $\times$ Treated</i>	-0.0892 (0.1421)	-0.0731*** (0.0218)	-0.2089 (0.1452)	-0.0701*** (0.0214)
<i>2020m8 $\times$ Treated</i>	-0.1262 (0.1431)	-0.0688*** (0.0172)	-0.2218* (0.1333)	-0.0663*** (0.0171)
<i>2020m9 $\times$ Treated</i>	-0.1693 (0.1394)	-0.0730*** (0.0169)	-0.2634** (0.1332)	-0.0703*** (0.0172)

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Table A6 – continued from previous page

	$\%Own.b, i$	$\mathbb{1}(\%Own.b, i > 0)$	$\%Own.b, i$	$\mathbb{1}(\%Own.b, i > 0)$
$2020m10 \times Treated$	-0.2424 (0.1490)	-0.0754*** (0.0179)	-0.2594* (0.1471)	-0.0750*** (0.0176)
$2020m11 \times Treated$	-0.3470** (0.1457)	-0.0790*** (0.0258)	-0.4549*** (0.1576)	-0.0764*** (0.0253)
$2020m12 \times Treated$	-0.2964* (0.1623)	-0.0827*** (0.0274)	-0.4335*** (0.1636)	-0.0789*** (0.0280)
$2021m1 \times Treated$	-0.3159** (0.1536)	-0.0950*** (0.0284)	-0.3740** (0.1468)	-0.0929*** (0.0282)
$2021m2 \times Treated$	-0.3325* (0.1739)	-0.1043*** (0.0279)	-0.3998** (0.1689)	-0.1019*** (0.0273)
$2021m3 \times Treated$	-0.2625 (0.1831)	-0.1203*** (0.0376)	-0.3501* (0.1787)	-0.1181*** (0.0378)
$2021m4 \times Treated$	-0.2455 (0.1555)	-0.0961** (0.0400)	-0.2182 (0.1965)	-0.0975** (0.0393)
$2021m5 \times Treated$	-0.1771 (0.2084)	-0.0751 (0.0508)	-0.2007 (0.2157)	-0.0746 (0.0507)
$2021m6 \times Treated$	-0.0123 (0.1721)	-0.0712 (0.0478)	-0.0449 (0.1676)	-0.0703 (0.0480)
$2021m7 \times Treated$	-0.1317 (0.1896)	-0.0727 (0.0441)	-0.1595 (0.1902)	-0.0711 (0.0435)
$2021m8 \times Treated$	-0.1308 (0.1875)	-0.0716 (0.0429)	-0.1763 (0.1808)	-0.0699 (0.0428)
$2021m9 \times Treated$	-0.0535 (0.1781)	-0.0546 (0.0403)	-0.1242 (0.1740)	-0.0530 (0.0401)
$\Delta_t O.S_b$	-0.5080* (0.2594)	-0.0926 (0.0551)	-0.4895** (0.2457)	-0.0928 (0.0554)
$FX \text{ to } EUR$	-2.0369 (2.5648)	0.0904 (0.4864)	-1.5076 (2.5710)	0.0659 (0.4936)
$L.Return_b^m$			-0.0004 (0.0010)	0.0000 (0.0002)
$L.Vol.Return_b^d$			0.0842* (0.0441)	-0.0025 (0.0042)
$post \ 2021m9 \times Treated$	-0.1427 (0.1658)	-0.0528* (0.0299)	-0.1554 (0.1622)	-0.0524* (0.0297)
$Constant$	0.8343 (2.4489)	0.3289 (0.4648)	0.1942 (2.4805)	0.3564 (0.4730)
Observations	191,074	238,656	191,074	238,656
adj.R ²		0.336		0.336
Banks	26	26	26	26
$Investor \times Month$ FE	X	X	X	X
$Bank$ FE	X	X	X	X

Figure A3**Event Study Plots: Funds and Institutions Unmatched Sample**

Panels A and B of Figure A3 plot the evolution of the coefficients $\{\delta_k\}$ of Equation (2). Panels A and B show the estimates using the poisson pseudo-maximum likelihood regression on $\%Ownership_{i,j,t}$ for funds and institutions on the unmatched sample. Panel C and D show the estimates using a linear probability model on an indicator variable being one if investor i is invested in bank b at time t for the unmatched sample. The sample includes 50 banks. The regression uses bank and $investor \times month$ fixed effects and controls for the exchange rate to EUR, $FX\ to\ EUR$, the monthly change in shares outstanding per bank $\Delta_t O.S._b$. The outer bands represent the sup-t 95% confidence bands according to Montiel Olea and Plagborg-Møller (2019), the inner bands are the 95% confidence intervals for clustered standard errors on banks. All coefficients measure the impact compared to February 2020. The average value of the dependent variable for the treated group in February 2020 is reported above the coefficient of February 2020. The dashed horizontal line uses January 2020 as a benchmark instead of February 2020. The dotted vertical line marks the month of the implementation, whereas the dashed vertical lines mark the different relief announcements.

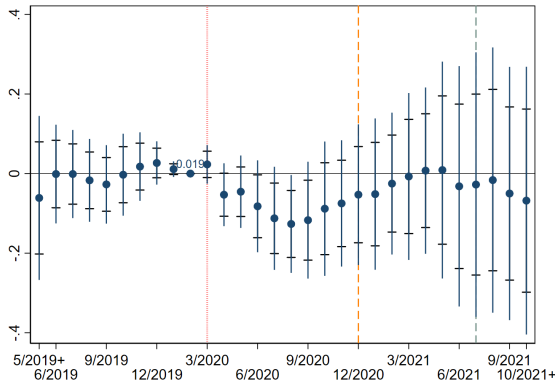
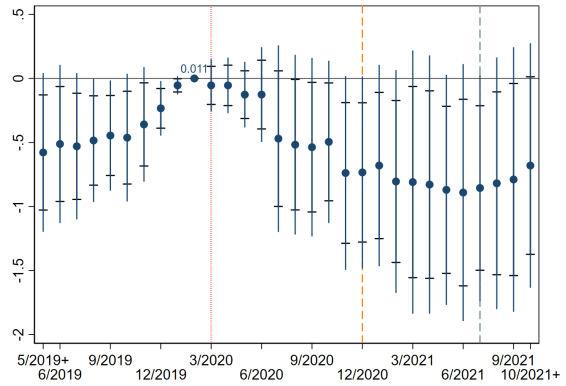
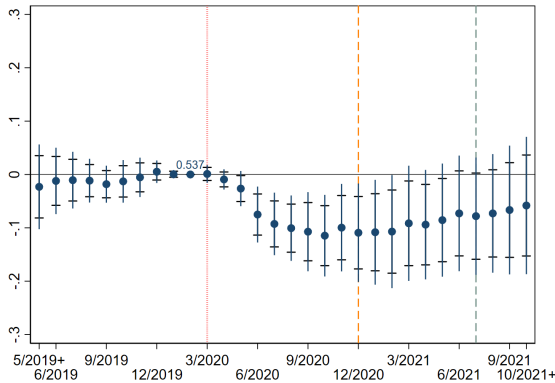
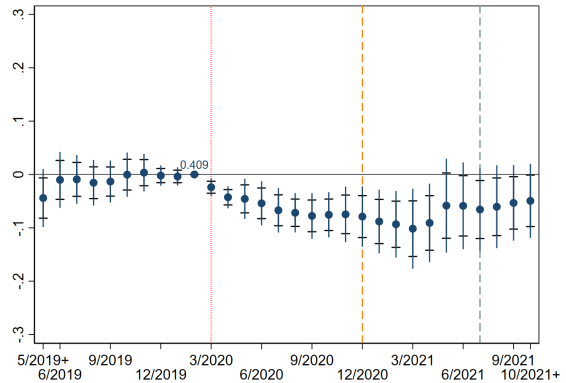
Panel A. Funds**Panel B. Institutions****Panel C. Funds LPM****Panel D. Institutions LPM**

Figure A4

Event Study Plots: ETFs and Pensions

Panels A and B of Figure A4 plot the evolution of the coefficients $\{\hat{\delta}_k\}$ of Equation (2) show the estimates using the poisson pseudo-maximum likelihood regression on $\%Ownership_{i,j,t}$ for ETFs and pension funds, respectively. Panel A shows the results for ETFs and Panel B shows the results for pension funds. The sample includes 26 banks. The regression uses bank and $investor \times month$ fixed effects and controls for the exchange rate to EUR, $FX\ to\ EUR$, and the monthly change in shares outstanding per bank $\Delta_t O.S.b.$. The outer bands represent the sup-t 95% confidence bands according to Montiel Olea and Plagborg-Møller (2019), the inner bands are the 95% confidence intervals for clustered standard errors on banks. All coefficients measure the impact compared to February 2020. The average value of the dependent variable for the treated group in February 2020 is reported above the coefficient of February 2020. The dotted red line marks the month of the implementation, the dashed orange line marks the relaxation announcement, and the dashed teal line marks the expiration announcement.

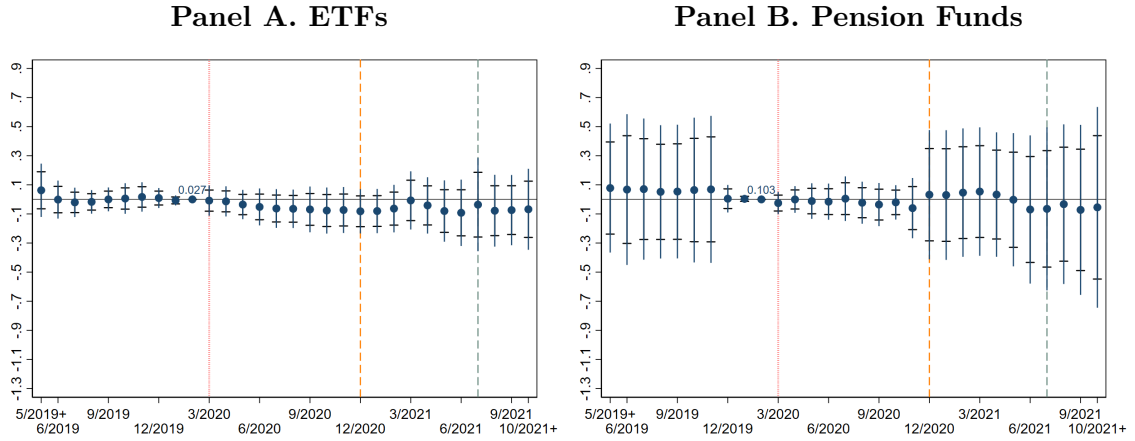
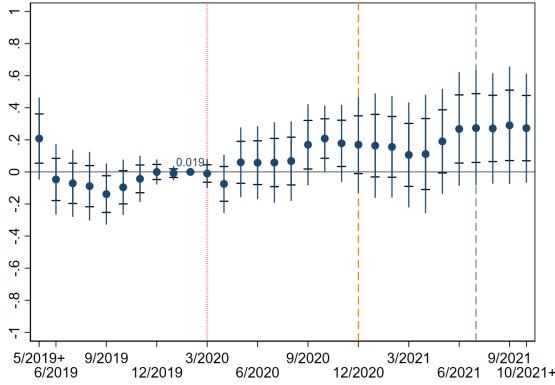


Figure A5

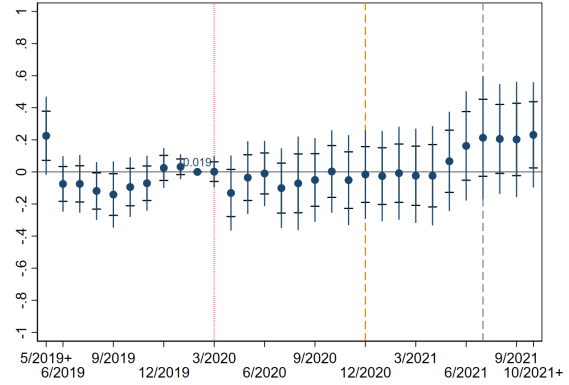
Event Study Plots Triple Difference: Funds Dividend Dimension

The baseline matched sample includes 26 banks. However, the effective number of banks plotted in each panel varies due to the availability of the respective characteristic variables used for the interaction (see *Table 6* for exact estimation counts). The regression uses bank and *investor* $\times$ *month* fixed effects and controls for the exchange rate to EUR, *FX to EUR*, the monthly change in shares outstanding per bank $\Delta_t O.S.b$, the monthly stock return of the previous month, and the banks' daily stock return volatility of the previous month. Panel A shows the results for the indicator variable of low RelVol 2018, Panel B shows the results for the indicator variable of low SOA in 2018, Panel C shows the results for the indicator variable of low expected 2020 dividend yield, and Panel D shows the results for the continuous variable of the expected 2020 dividend yield. The outer bands represent the sup-t 95% confidence bands according to Montiel Olea and Plagborg-Møller (2019), the inner bands are the 95% confidence intervals for clustered standard errors on banks. All coefficients measure the impact compared to February 2020. The average value of the dependent variable for the treated group in February 2020 is reported above the coefficient of February 2020. The dotted red line marks the month of the implementation, the dashed orange line marks the relaxation announcement, and the dashed teal line marks the expiration announcement.

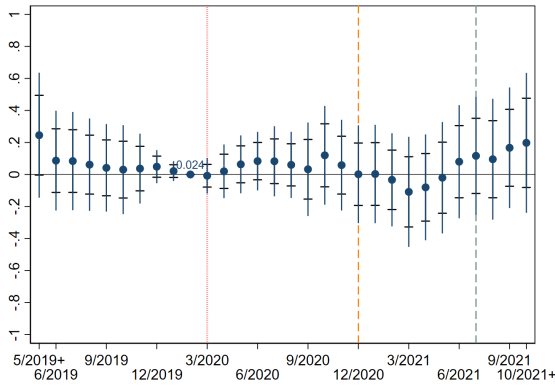
Panel A. RelVol



Panel B. SOA



Panel C. Div. Yield



Panel D. Div. Yield Continuous

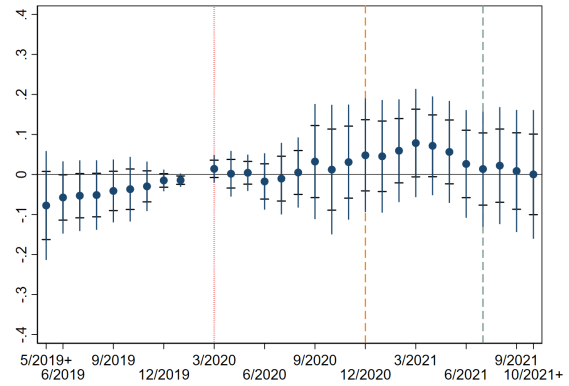
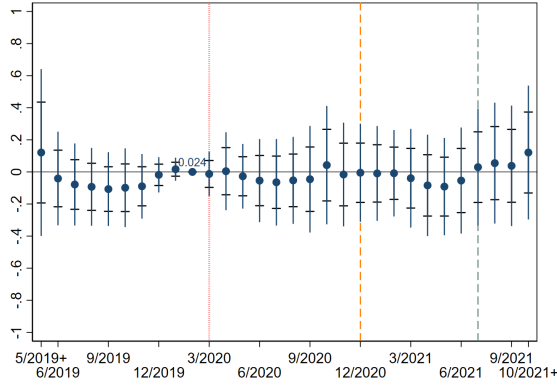


Figure A6

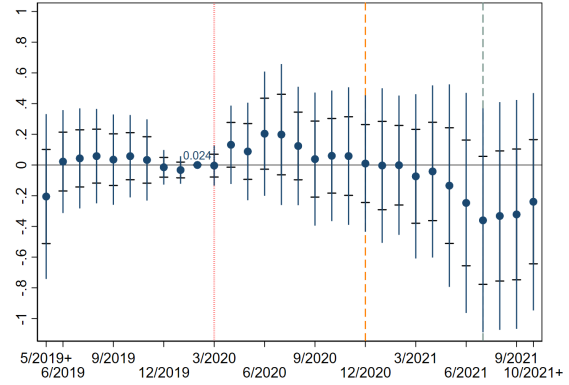
Event Study Plots Triple Difference: Funds Risk Dimension

The baseline matched sample includes 26 banks. However, the effective number of banks plotted in each panel varies due to the availability of the respective characteristic variables used for the interaction (see *Table ??* for exact estimation counts). The regression uses bank and *investor* $\times$ *month* fixed effects and controls for the exchange rate to EUR, *FX to EUR*, the monthly change in shares outstanding per bank $\Delta_t O.S._b$, the monthly stock return of the previous month, and the banks' daily stock return volatility of the previous month. Panel A shows the results for the indicator variable of low Price/Book values 2019, Panel B shows the results for the indicator variable of low Log(Z-Score) in 2019, Panel C shows the results for the indicator variable of low Tier 1 Cap. Ratio in 2019. The outer bands represent the sup-t 95% confidence bands according to Montiel Olea and Plagborg-Møller (2019), the inner bands are the 95% confidence intervals for clustered standard errors on banks. All coefficients measure the impact compared to February 2020. The average value of the dependent variable for the treated group in February 2020 is reported above the coefficient of February 2020. The dotted red line marks the month of the implementation, the dashed orange line marks the relaxation announcement, and the dashed teal line marks the expiration announcement.

Panel A. Price/Book



Panel B. Log(Z-Score)



Panel C. Tier 1 Cap. Ratio

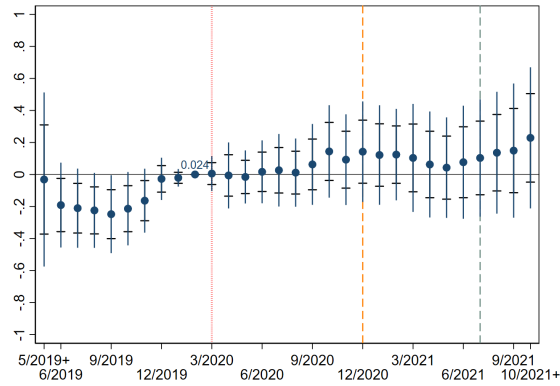


Table A7

CAR Regressions Additional Controls: Event Days

Table A7 presents the regressions of equation 9 for the three events under study using the matched sample of 26 banks. Each column shows the CAR for a different event date where the event window is set to $[-1, 1]$ and the start date is displayed in the first row. Abnormal returns are calculated using the Fama-French 3 factor model. The regression controls for the market value, *MktValue*, and the dividend yield of the year before the restrictions, *DivYield2019*. Standard errors are robust. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively.

	Restriction	Relaxation				Expiration		
Event Window Start:	03/27/20	11/04/20	11/24/20	12/02/20	12/15/20	06/14/21	06/30/21	07/22/21
Panel A: Treatment Effects								
<i>Treated</i>	-1.047 (1.638)	-0.736 (0.918)	-1.309 (1.324)	0.628 (1.287)	1.167 (0.824)	-0.665 (0.663)	1.067 (0.727)	2.134** (0.772)
<i>DivYield2019</i>	-0.359 (0.268)	0.154 (0.223)	-0.092 (0.241)	-0.392 (0.364)	-0.051 (0.140)	0.131 (0.185)	-0.021 (0.108)	-0.020 (0.206)
<i>Constant</i>	1.027 (1.434)	-0.671 (1.117)	0.136 (1.169)	1.476 (1.285)	-0.754 (0.603)	-0.016 (0.691)	-0.472 (0.619)	-0.244 (0.768)
N	26	26	26	26	26	26	26	26
adj. R^2	-0.016	-0.082	-0.078	0.036	0.028	-0.053	-0.036	0.117
Controls	X	X	X	X	X	X	X	X
Panel B: Fund Share								
<i>Treated</i>	1.945 (2.380)	-1.157 (1.104)	-0.084 (2.085)	2.858 (1.946)	2.488** (1.188)	-0.338 (0.952)	1.492 (1.306)	3.242*** (0.986)
<i>High Fund Share</i>	3.667 (2.710)	-0.287 (2.032)	-3.010* (1.468)	0.411 (0.958)	2.140* (1.050)	0.123 (0.791)	1.283 (0.777)	2.575** (0.938)
<i>Treated</i> $\times$ <i>High Fund Share</i>	-7.242* (3.806)	0.931 (2.172)	-0.993 (2.461)	-4.419** (2.023)	-3.457* (1.686)	-0.679 (1.281)	-1.313 (1.421)	-3.143** (1.301)
<i>DivYield2019</i>	-0.074 (0.407)	0.105 (0.239)	0.142 (0.195)	-0.111 (0.293)	0.069 (0.171)	0.170 (0.205)	-0.002 (0.113)	0.052 (0.231)
<i>Constant</i>	-0.980 (1.606)	-0.409 (1.182)	-0.266 (1.168)	0.199 (1.028)	-1.695** (0.678)	-0.210 (0.798)	-0.842 (0.752)	-1.127 (0.807)
N	26	26	26	26	26	26	26	26
adj. R^2	0.032	-0.176	0.050	0.156	0.048	-0.137	-0.099	0.146
Controls	X	X	X	X	X	X	X	X

Table A8
CAR Regressions Continuous Fund Ownership: Event Days

Table A8 presents the regressions of equation 9 for the three events under study using the matched sample of 26 banks. The indicator variable has been replaced by the fund ownership in percentage points in the previous month. Each column shows the CAR for a different event date where the event window is set to $[-1, 1]$ and the start date is displayed in the first row. Abnormal returns are calculated using the Fama-French 3 factor model. The regression controls for the market value, *MktValue*. Standard errors are robust. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively.

	Restriction	Relaxation				Expiration		
Event Window Start:	03/27/20	11/04/20	11/24/20	12/02/20	12/15/20	06/14/21	06/30/21	07/22/21
Treated	3.961 (2.439)	-0.231 (1.714)	0.481 (2.273)	5.300** (2.380)	2.377* (1.378)	-0.344 (1.393)	1.450 (1.282)	2.579* (1.373)
% of Funds _{<i>t</i>-1}	0.180* (0.102)	-0.031 (0.079)	-0.130** (0.059)	0.022 (0.041)	0.100** (0.040)	-0.006 (0.025)	0.041 (0.026)	0.111*** (0.039)
Treated $\times$ % of Funds _{<i>t</i>-1}	-0.301** (0.117)	0.004 (0.094)	-0.008 (0.080)	-0.211** (0.079)	-0.100* (0.050)	-0.006 (0.052)	-0.034 (0.039)	-0.065 (0.054)
Constant	-2.067** (0.825)	0.022 (1.033)	0.897 (1.023)	-0.222 (0.400)	-1.901*** (0.383)	0.410 (0.456)	-0.961 (0.680)	-1.401*** (0.319)
N	26	26	26	26	26	26	26	26
r2_a	0.077	-0.143	0.107	0.292	0.066	-0.145	-0.058	0.217
date	03/27/20	11/04/20	11/24/20	12/02/20	12/15/20	06/14/21	06/30/21	07/22/21

Table A9
CAR Regressions 5 Factor Model Additional Controls: Event Days

Table A9 presents the regressions of equation 9 for the three events under study using the matched sample of 26 banks. Each column shows the CAR for a different event date where the event window is set to $[-1, 1]$ and the start date is displayed in the first row. Abnormal returns are calculated using the Fama-French 5 factor model. The regression controls for the market value, *MktValue*, and the dividend yield of the year before the restrictions, *DivYield2019*. Standard errors are robust. ***, **, and * indicate statistically different from zero at the 1%, 5%, and 10% level of significance, respectively.

	Restriction	Relaxation				Expiration		
Event Window Start:	03/27/20	11/04/20	11/24/20	12/02/20	12/15/20	06/14/21	06/30/21	07/22/21
Panel A: Treatment Effects								
<i>Treated</i>	-0.321 (1.616)	-0.888 (1.066)	-1.468 (1.308)	1.189 (1.160)	0.811 (0.867)	-0.447 (0.683)	1.365* (0.735)	1.990** (0.808)
<i>DivYield2019</i>	-0.155 (0.210)	0.375** (0.173)	-0.153 (0.248)	-0.440 (0.331)	0.013 (0.137)	0.118 (0.194)	-0.033 (0.110)	0.002 (0.213)
N	26	26	26	26	26	26	26	26
adj. R ²	0.022	0.053	-0.061	0.039	0.087	-0.074	0.008	0.070
Panel B: Fund Share								
<i>Treated</i>	3.647* (1.771)	-0.520 (1.510)	-0.457 (2.218)	3.088* (1.732)	2.357* (1.345)	-0.108 (0.984)	1.755 (1.315)	3.251*** (1.071)
<i>High Fund Share</i>	2.522 (2.719)	-2.137 (1.928)	-2.500 (1.473)	0.708 (1.088)	1.627 (0.986)	-0.188 (0.704)	1.210 (0.797)	2.664*** (0.902)
<i>Treated × High Fund Share</i>	-8.610** (3.388)	0.298 (2.266)	-0.812 (2.591)	-3.923** (1.832)	-3.656* (1.790)	-0.582 (1.243)	-1.216 (1.431)	-3.472** (1.339)
<i>DivYield2019</i>	0.286 (0.355)	0.474** (0.181)	0.041 (0.215)	-0.210 (0.267)	0.175 (0.171)	0.167 (0.210)	-0.017 (0.114)	0.091 (0.234)
N	26	26	26	26	26	26	26	26
adj. R ²	0.238	0.054	-0.013	0.116	0.130	-0.147	-0.057	0.110
Controls	X	X	X	X	X	X	X	X

Figure A7
CARs and Fund Ownership

Figure A7 shows the residualized event date cumulative abnormal returns in percentage points for the events dates used in *Table 8* of treated banks plotted against residualized lagged fund ownership for each announcement date considered in Table 8. Both variables are residualized with respect to banks' market value. The green dots represent treated banks, while the orange line shows the fitted relationship estimated for the control banks, providing the benchmark against which the treated observations can be compared.

